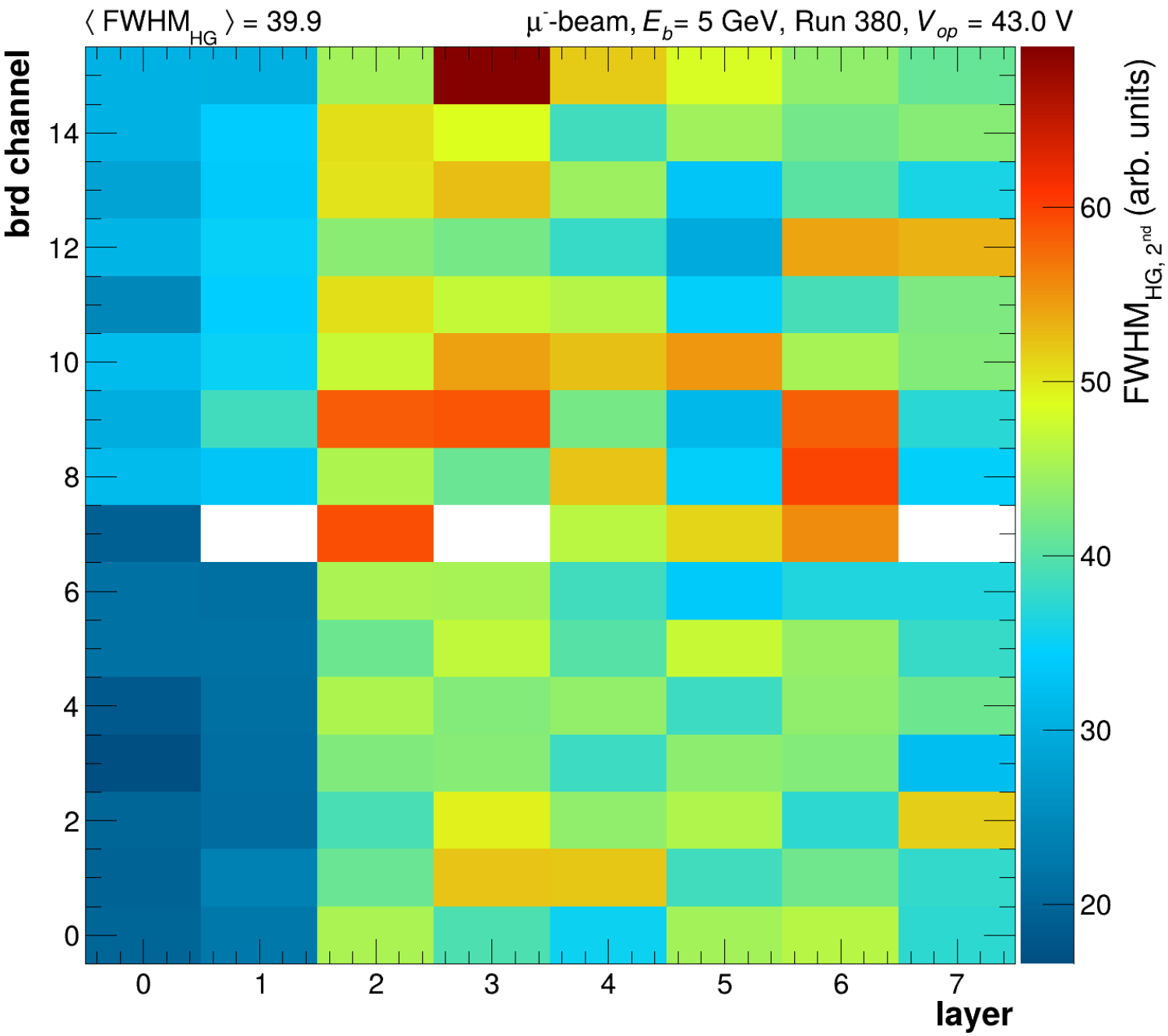
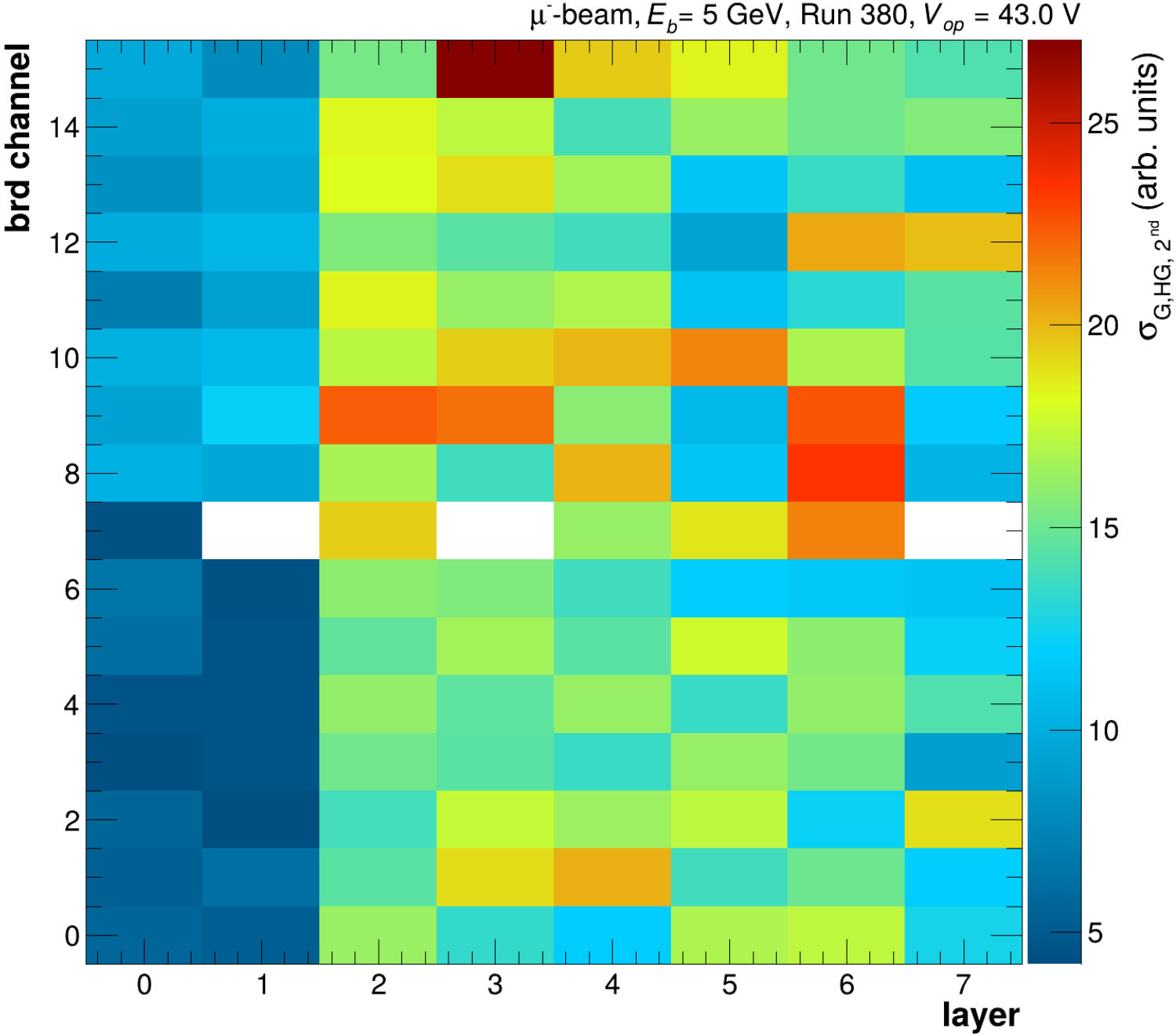
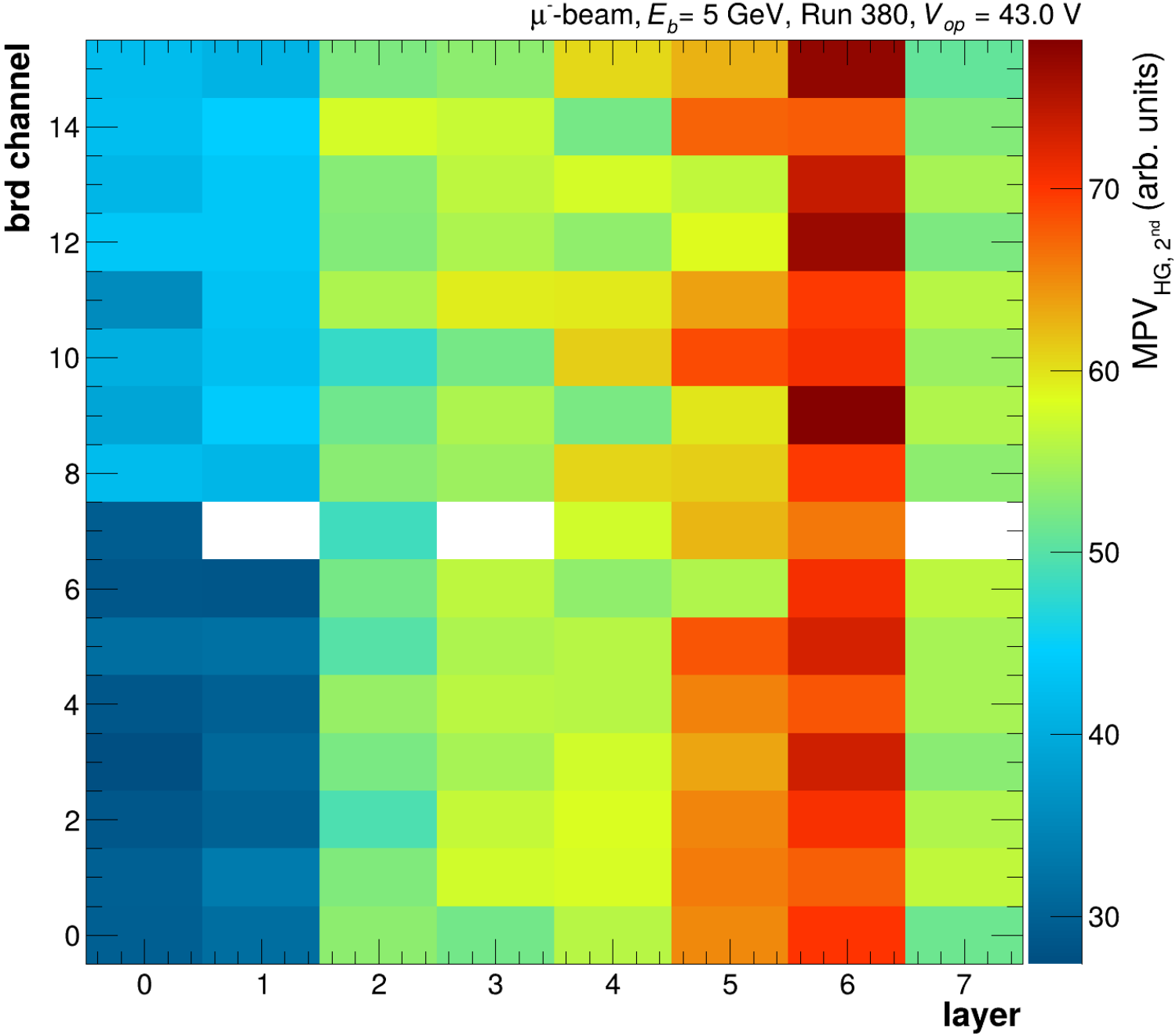
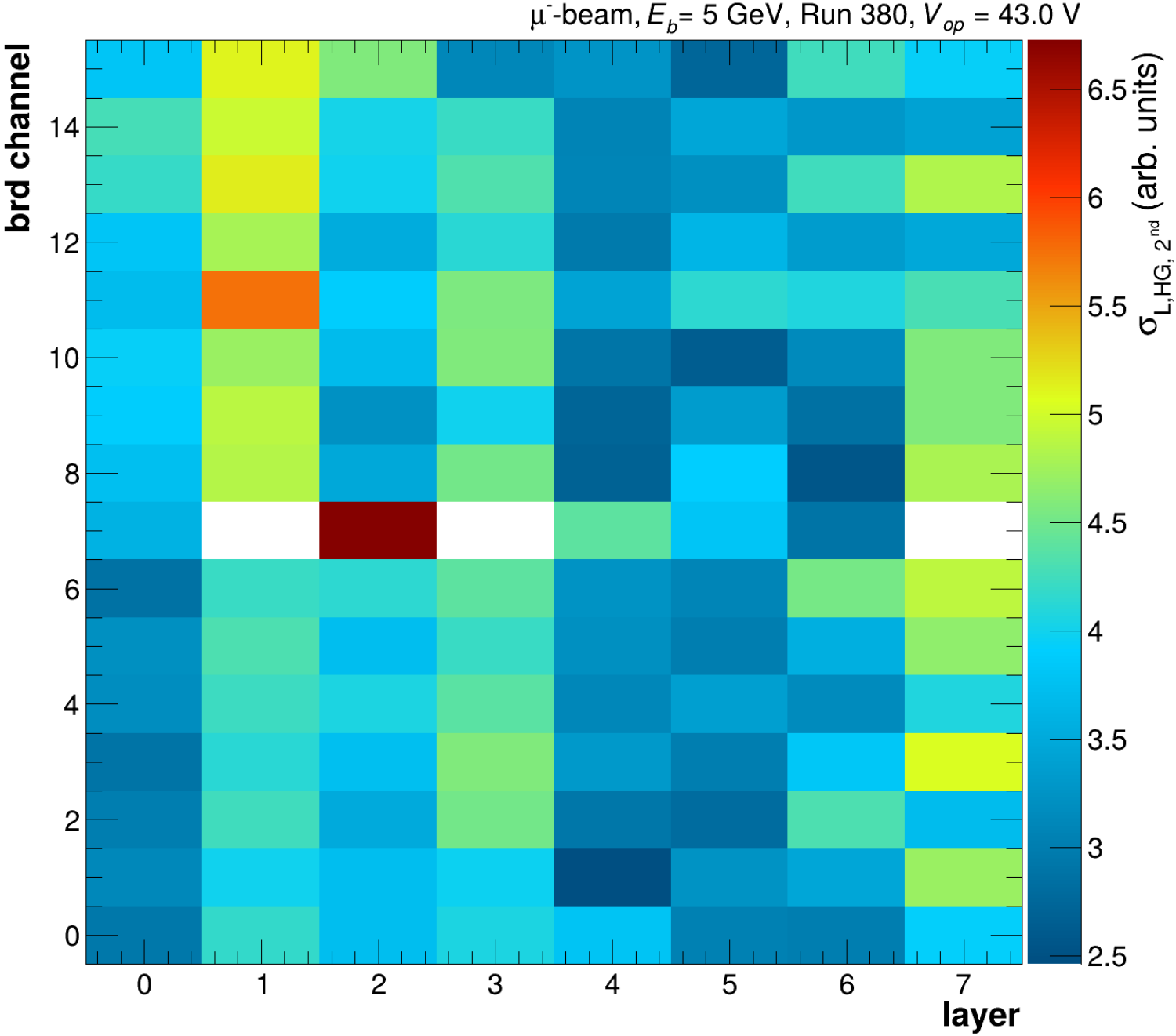
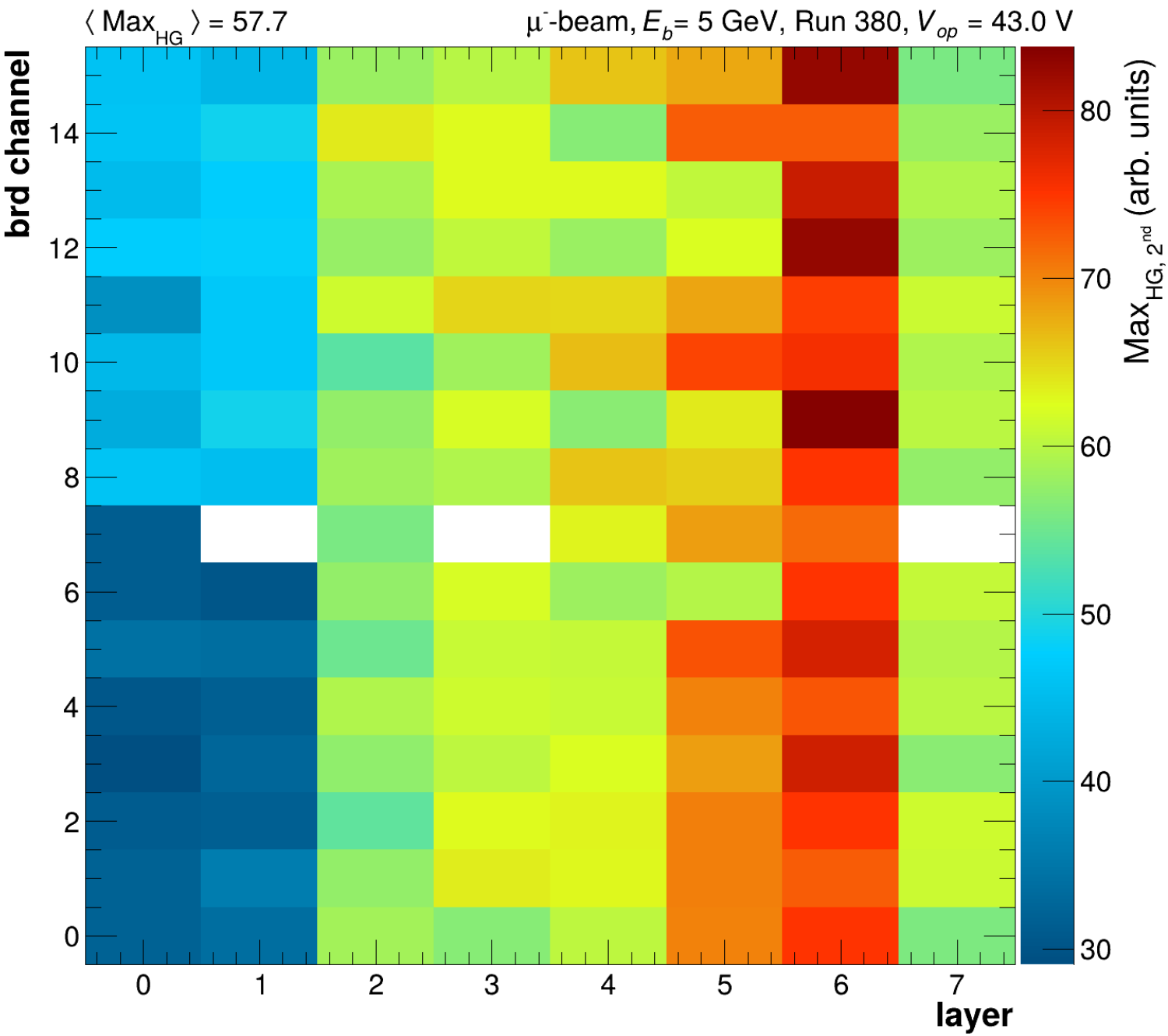
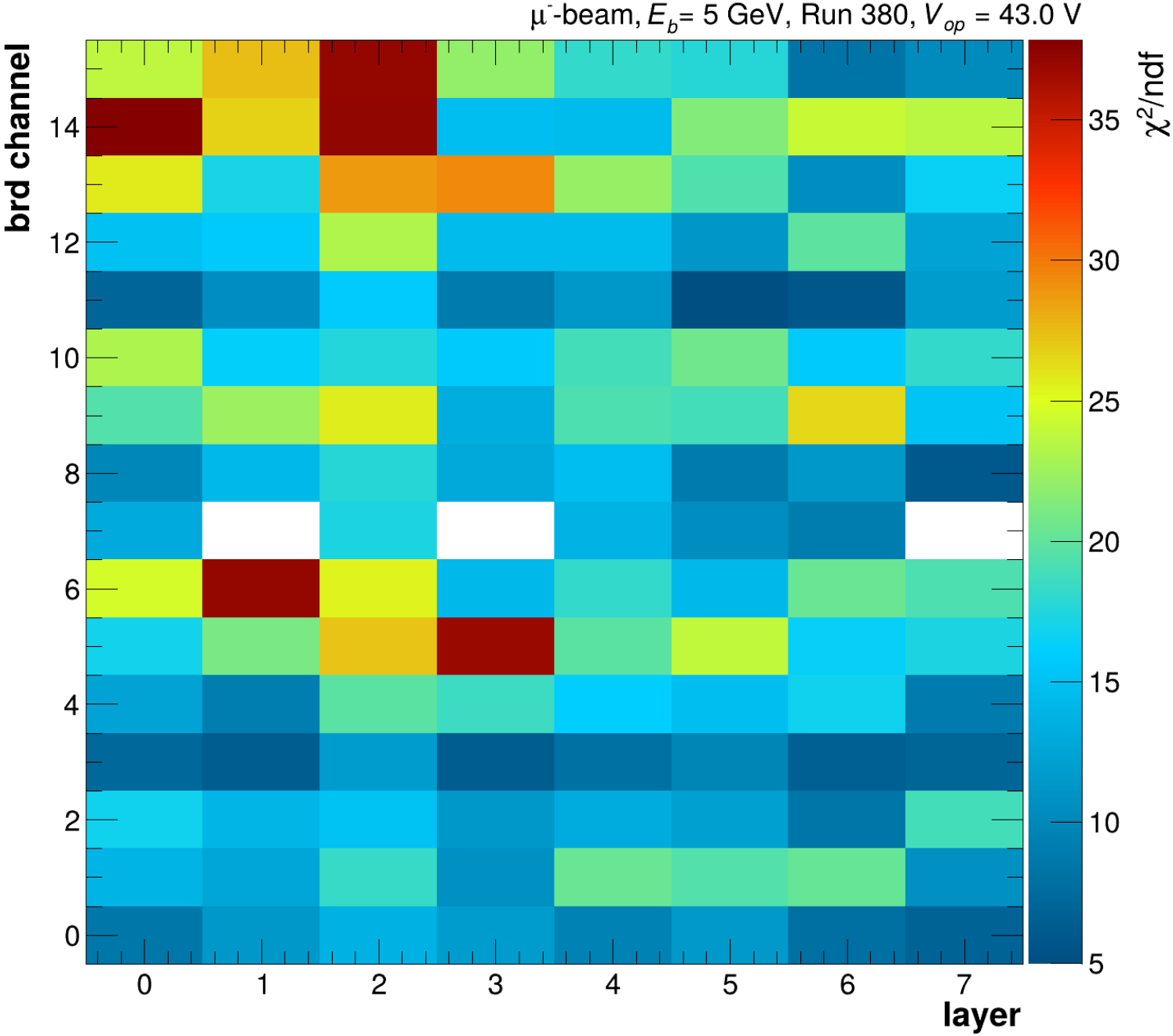

⟨ FWHM$_{HG}$ ⟩ = 39.9
μ$^-$-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
layer
FWHM$_{HG,\ 2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
30
40
50
60

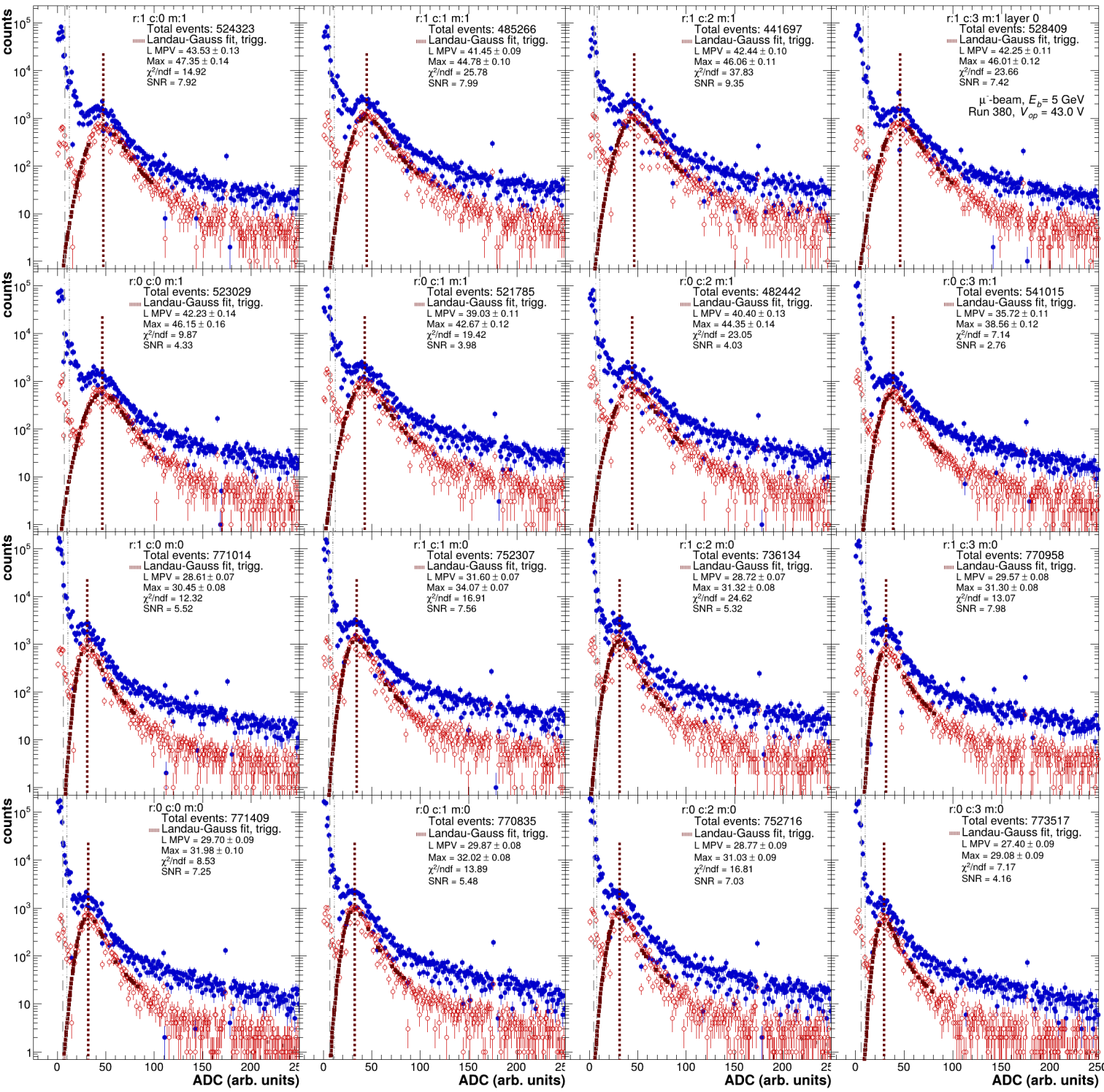

μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
5
10
15
20
25
$\sigma_{G,HG,2^{nd}}$ (arb. units)

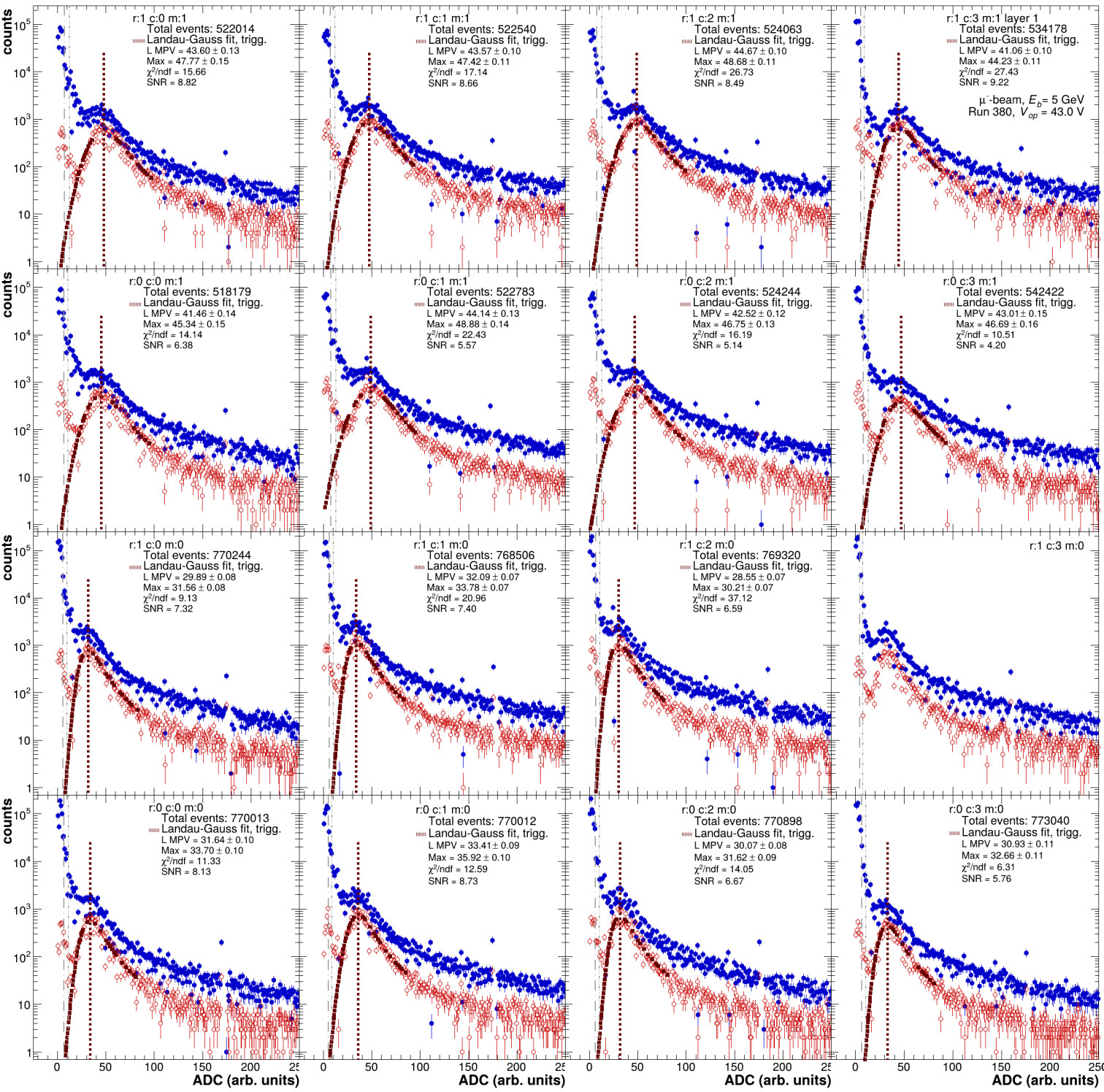

μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
MPV$_{HG,\,2^{nd}}$ (arb. units)
30
40
50
60
70

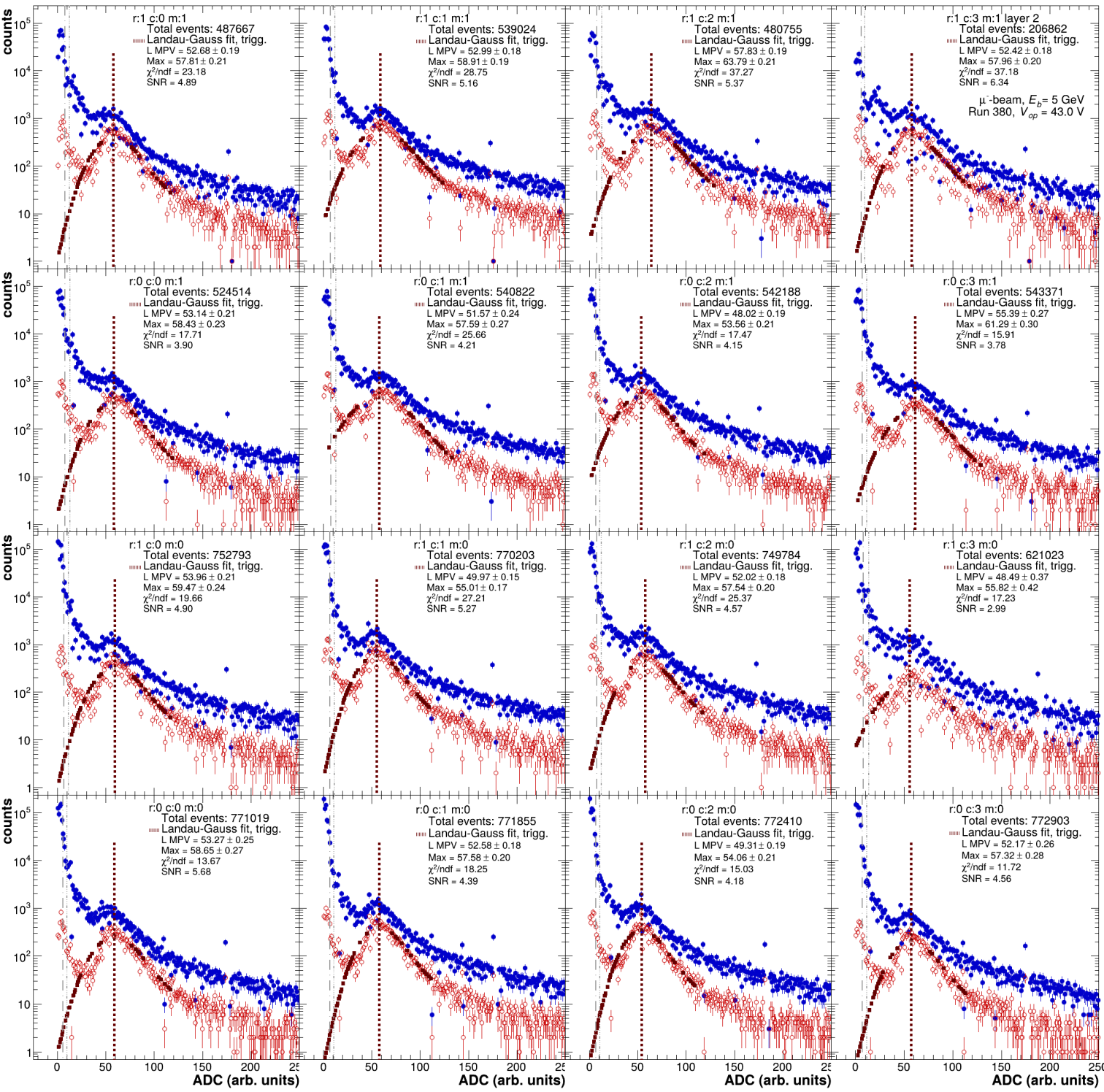

μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
2.5
3
3.5
4
4.5
5
5.5
6
6.5

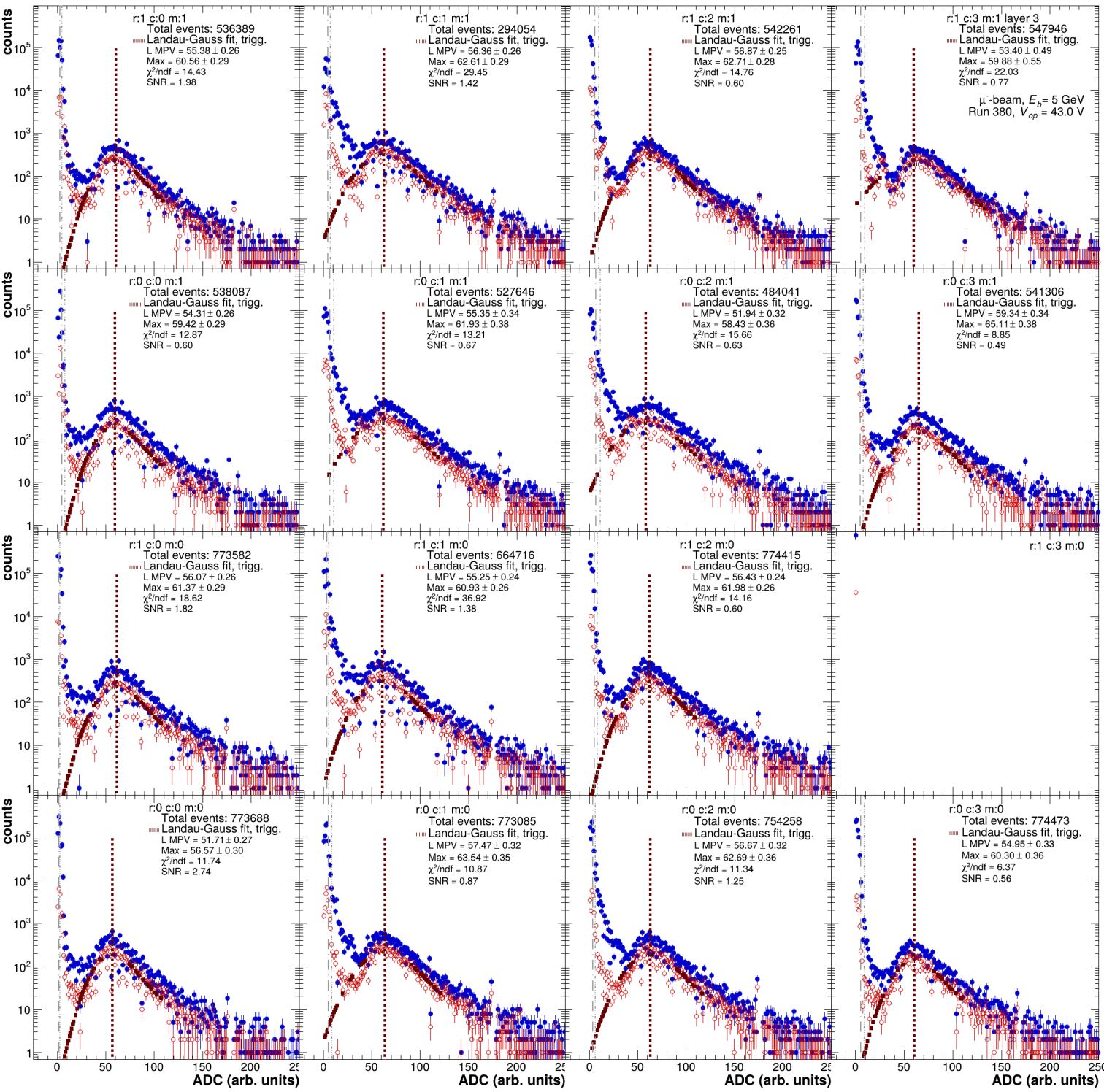

⟨ Max_HG ⟩ = 57.7
μ⁻-beam, E_b = 5 GeV, Run 380, V_op = 43.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
30
40
50
60
70
80

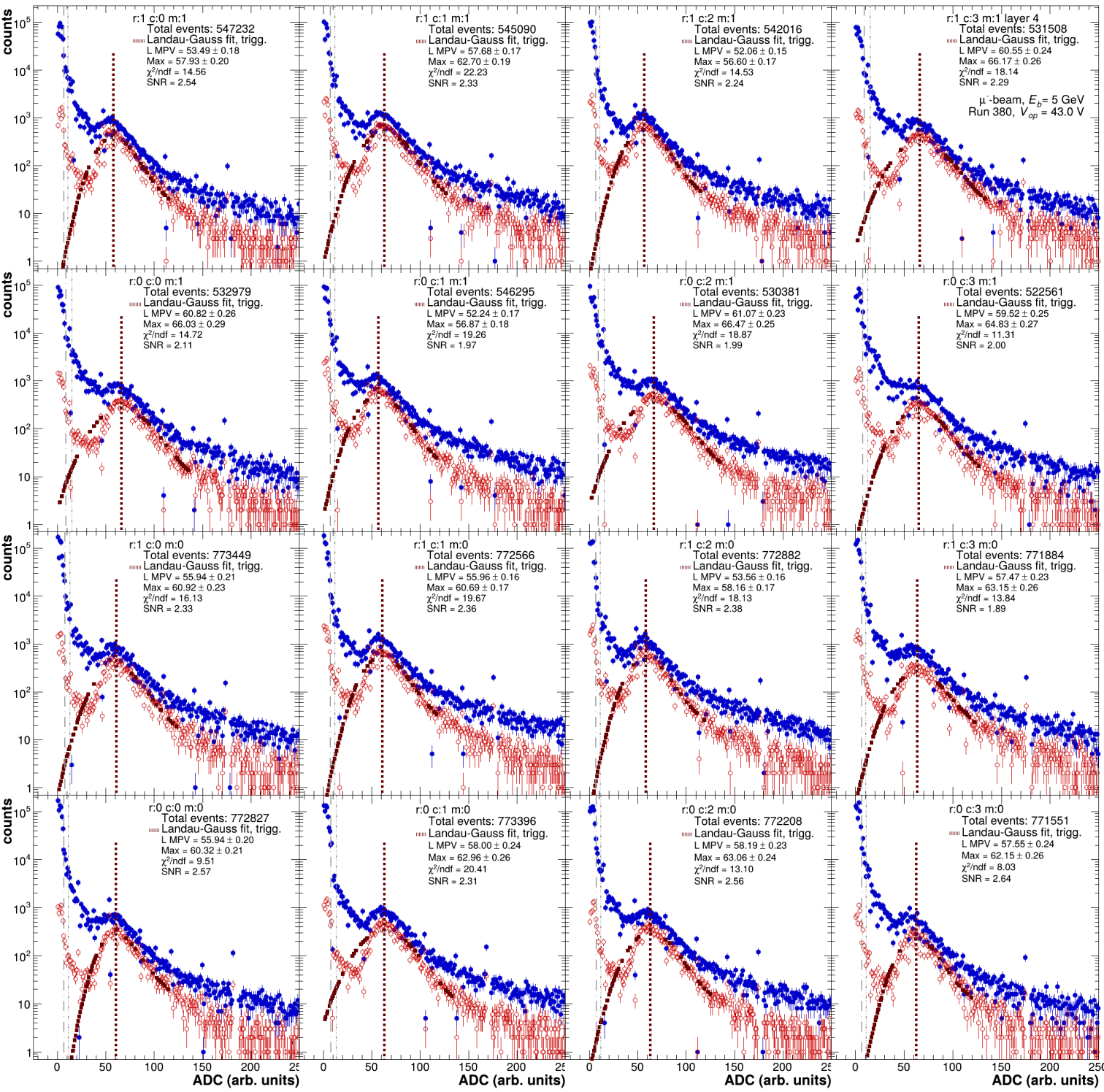

μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
χ^2/ndf
5
10
15
20
25
30
35

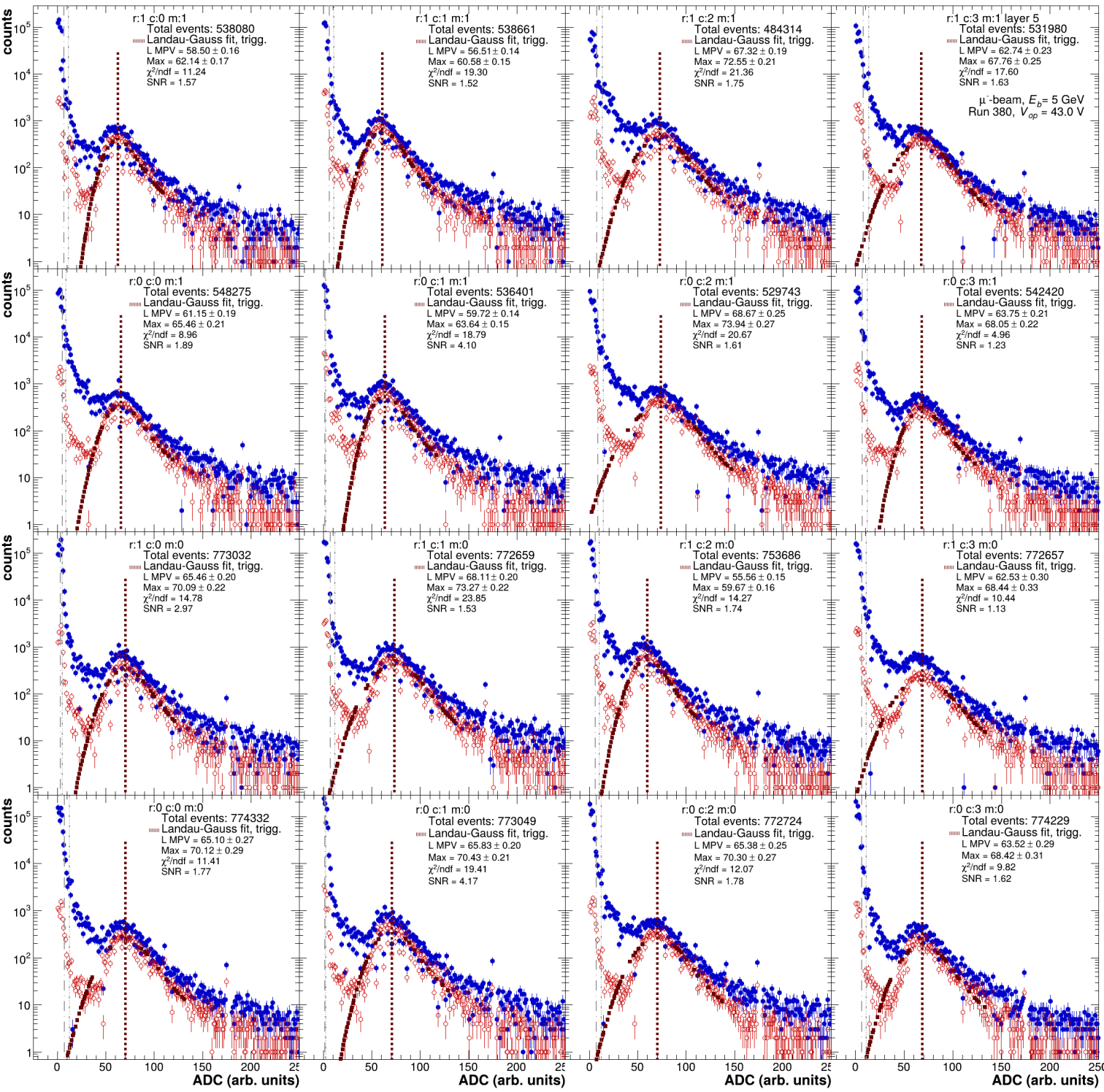

r:1 c:0 m:1
Total events: 524323
Landau-Gauss fit, trigg.
L MPV = 43.53 ± 0.13
Max = 47.35 ± 0.14
χ²/ndf = 14.92
SNR = 7.92
r:1 c:1 m:1
Total events: 485266
Landau-Gauss fit, trigg.
L MPV = 41.45 ± 0.09
Max = 44.78 ± 0.10
χ²/ndf = 25.78
SNR = 7.99
r:1 c:2 m:1
Total events: 441697
Landau-Gauss fit, trigg.
L MPV = 42.44 ± 0.10
Max = 46.06 ± 0.11
χ²/ndf = 37.83
SNR = 9.35
r:1 c:3 m:1 layer 0
Total events: 528409
Landau-Gauss fit, trigg.
L MPV = 42.25 ± 0.11
Max = 46.01 ± 0.12
χ²/ndf = 23.66
SNR = 7.42
μ⁻-beam, E_b = 5 GeV
Run 380, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 523029
Landau-Gauss fit, trigg.
L MPV = 42.23 ± 0.14
Max = 46.15 ± 0.16
χ²/ndf = 9.87
SNR = 4.33
r:0 c:1 m:1
Total events: 521785
Landau-Gauss fit, trigg.
L MPV = 39.03 ± 0.11
Max = 42.67 ± 0.12
χ²/ndf = 19.42
SNR = 3.98
r:0 c:2 m:1
Total events: 482442
Landau-Gauss fit, trigg.
L MPV = 40.40 ± 0.13
Max = 44.35 ± 0.14
χ²/ndf = 23.05
SNR = 4.03
r:0 c:3 m:1
Total events: 541015
Landau-Gauss fit, trigg.
L MPV = 35.72 ± 0.11
Max = 38.56 ± 0.12
χ²/ndf = 7.14
SNR = 2.76
r:1 c:0 m:0
Total events: 771014
Landau-Gauss fit, trigg.
L MPV = 28.61 ± 0.07
Max = 30.45 ± 0.08
χ²/ndf = 12.32
SNR = 5.52
r:1 c:1 m:0
Total events: 752307
Landau-Gauss fit, trigg.
L MPV = 31.60 ± 0.07
Max = 34.07 ± 0.07
χ²/ndf = 16.91
SNR = 7.56
r:1 c:2 m:0
Total events: 736134
Landau-Gauss fit, trigg.
L MPV = 28.72 ± 0.07
Max = 31.32 ± 0.08
χ²/ndf = 24.62
SNR = 5.32
r:1 c:3 m:0
Total events: 770958
Landau-Gauss fit, trigg.
L MPV = 29.57 ± 0.08
Max = 31.30 ± 0.08
χ²/ndf = 13.07
SNR = 7.98
r:0 c:0 m:0
Total events: 771409
Landau-Gauss fit, trigg.
L MPV = 29.70 ± 0.09
Max = 31.98 ± 0.10
χ²/ndf = 8.53
SNR = 7.25
r:0 c:1 m:0
Total events: 770835
Landau-Gauss fit, trigg.
L MPV = 29.87 ± 0.08
Max = 32.02 ± 0.08
χ²/ndf = 13.89
SNR = 5.48
r:0 c:2 m:0
Total events: 752716
Landau-Gauss fit, trigg.
L MPV = 28.77 ± 0.09
Max = 31.03 ± 0.09
χ²/ndf = 16.81
SNR = 7.03
r:0 c:3 m:0
Total events: 773517
Landau-Gauss fit, trigg.
L MPV = 27.40 ± 0.09
Max = 29.08 ± 0.09
χ²/ndf = 7.17
SNR = 4.16
counts
ADC (arb. units)

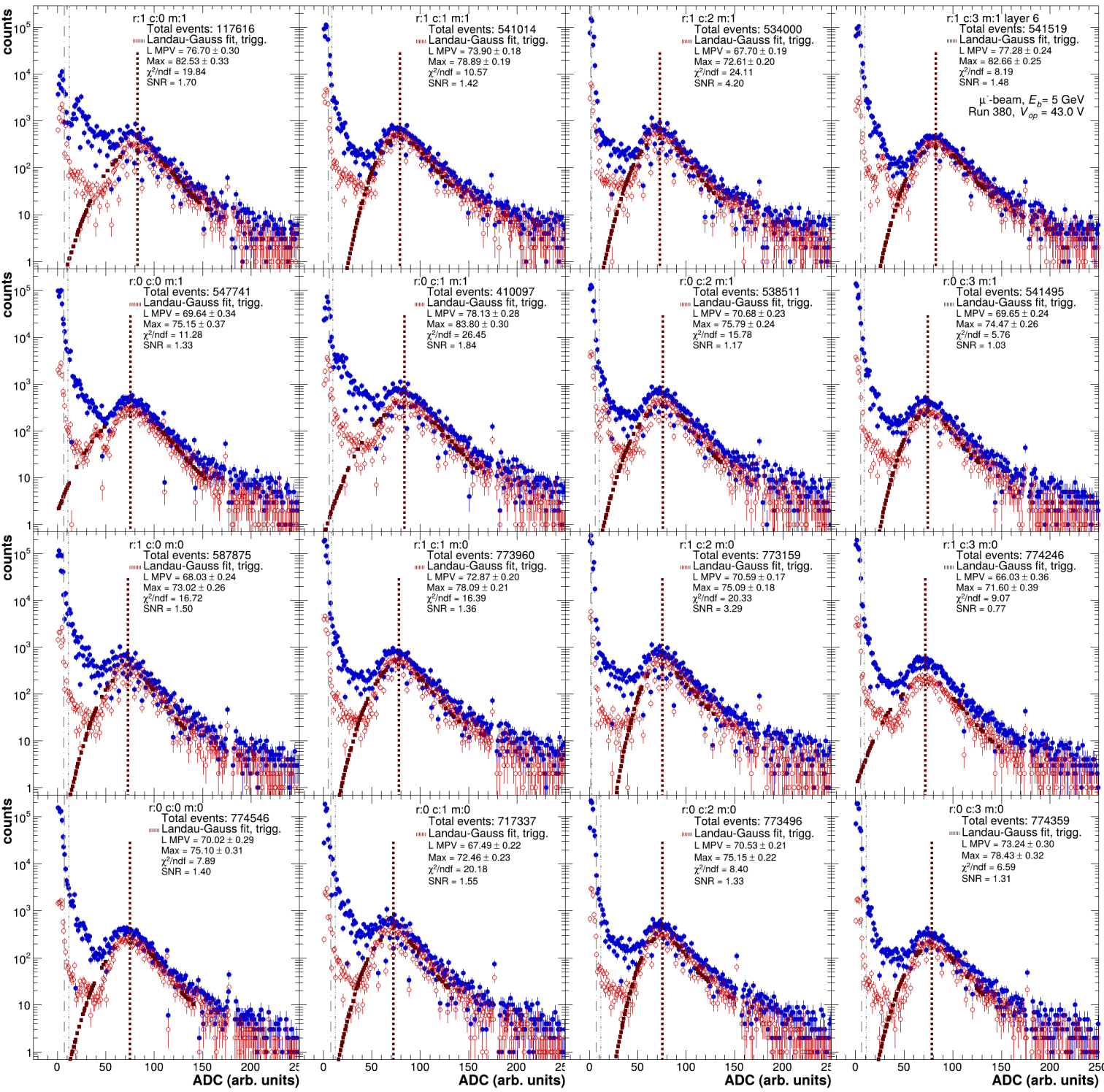

r:1 c:0 m:1
Total events: 522014
Landau-Gauss fit, trigg.
L MPV = 43.60 ± 0.13
Max = 47.77 ± 0.15
χ²/ndf = 15.66
SNR = 8.82
r:1 c:1 m:1
Total events: 522540
Landau-Gauss fit, trigg.
L MPV = 43.57 ± 0.10
Max = 47.42 ± 0.11
χ²/ndf = 17.14
SNR = 8.66
r:1 c:2 m:1
Total events: 524063
Landau-Gauss fit, trigg.
L MPV = 44.67 ± 0.10
Max = 48.68 ± 0.11
χ²/ndf = 26.73
SNR = 8.49
r:1 c:3 m:1 layer 1
Total events: 534178
Landau-Gauss fit, trigg.
L MPV = 41.06 ± 0.10
Max = 44.23 ± 0.11
χ²/ndf = 27.43
SNR = 9.22
μ-beam, E_b = 5 GeV
Run 380, V_op = 43.0 V
r:0 c:0 m:1
Total events: 518179
Landau-Gauss fit, trigg.
L MPV = 41.46 ± 0.14
Max = 45.34 ± 0.15
χ²/ndf = 14.14
SNR = 6.38
r:0 c:1 m:1
Total events: 522783
Landau-Gauss fit, trigg.
L MPV = 44.14 ± 0.13
Max = 48.88 ± 0.14
χ²/ndf = 22.43
SNR = 5.57
r:0 c:2 m:1
Total events: 524244
Landau-Gauss fit, trigg.
L MPV = 42.52 ± 0.12
Max = 46.75 ± 0.13
χ²/ndf = 16.19
SNR = 5.14
r:0 c:3 m:1
Total events: 542422
Landau-Gauss fit, trigg.
L MPV = 43.01 ± 0.15
Max = 46.69 ± 0.16
χ²/ndf = 10.51
SNR = 4.20
r:1 c:0 m:0
Total events: 770244
Landau-Gauss fit, trigg.
L MPV = 29.89 ± 0.08
Max = 31.56 ± 0.08
χ²/ndf = 9.13
SNR = 7.32
r:1 c:1 m:0
Total events: 768506
Landau-Gauss fit, trigg.
L MPV = 32.09 ± 0.07
Max = 33.78 ± 0.07
χ²/ndf = 20.96
SNR = 7.40
r:1 c:2 m:0
Total events: 769320
Landau-Gauss fit, trigg.
L MPV = 28.55 ± 0.07
Max = 30.21 ± 0.07
χ²/ndf = 37.12
SNR = 6.59
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 770013
Landau-Gauss fit, trigg.
L MPV = 31.64 ± 0.10
Max = 33.70 ± 0.10
χ²/ndf = 11.33
SNR = 8.13
r:0 c:1 m:0
Total events: 770012
Landau-Gauss fit, trigg.
L MPV = 33.41 ± 0.09
Max = 35.92 ± 0.10
χ²/ndf = 12.59
SNR = 8.73
r:0 c:2 m:0
Total events: 770898
Landau-Gauss fit, trigg.
L MPV = 30.07 ± 0.08
Max = 31.62 ± 0.09
χ²/ndf = 14.05
SNR = 6.67
r:0 c:3 m:0
Total events: 773040
Landau-Gauss fit, trigg.
L MPV = 30.93 ± 0.11
Max = 32.66 ± 0.11
χ²/ndf = 6.31
SNR = 5.76
counts
ADC (arb. units)

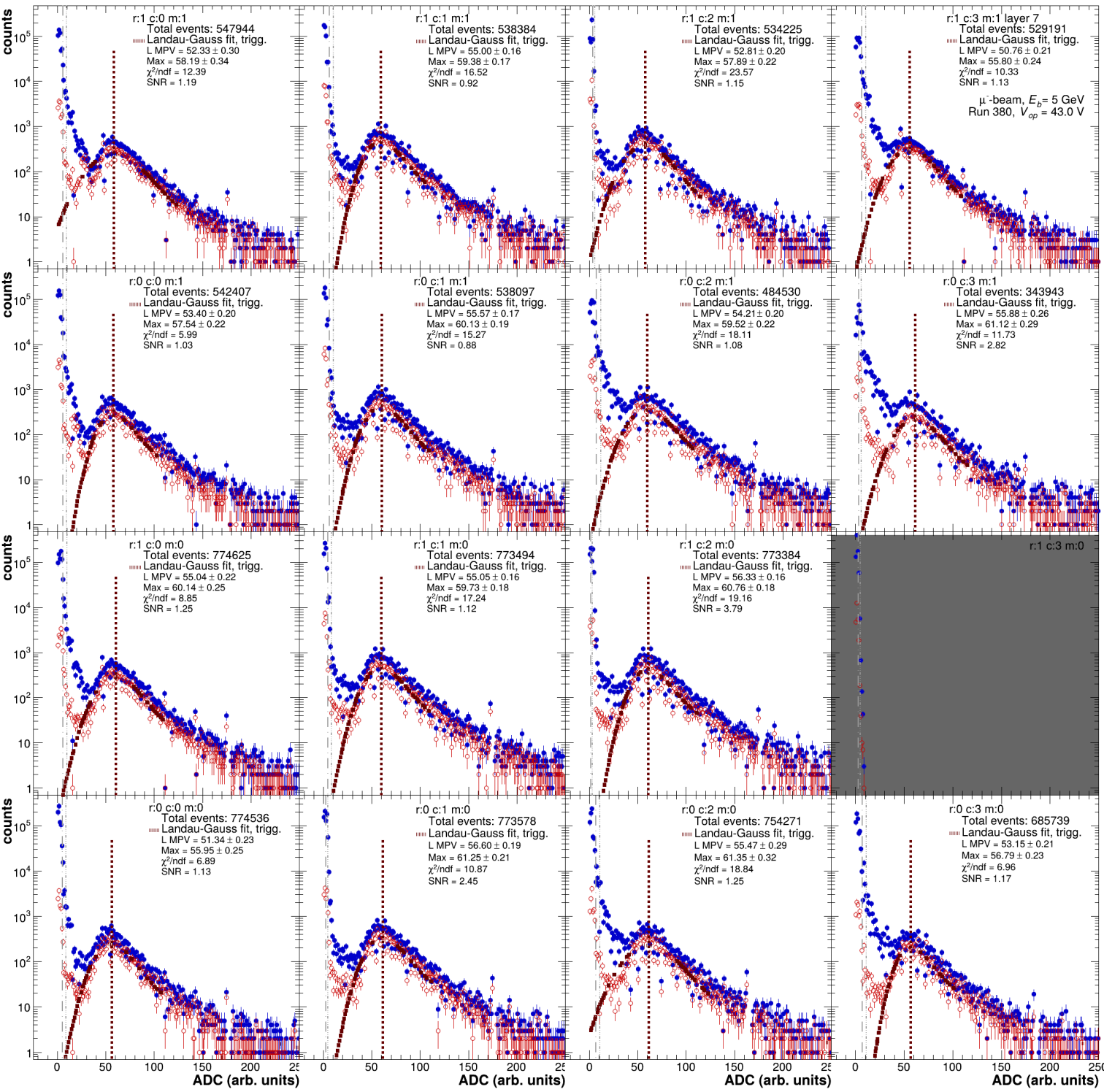

r:1 c:0 m:1
Total events: 487667
Landau-Gauss fit, trigg.
L MPV = 52.68 ± 0.19
Max = 57.81 ± 0.21
χ²/ndf = 23.18
SNR = 4.89
r:1 c:1 m:1
Total events: 539024
Landau-Gauss fit, trigg.
L MPV = 52.99 ± 0.18
Max = 58.91 ± 0.19
χ²/ndf = 28.75
SNR = 5.16
r:1 c:2 m:1
Total events: 480755
Landau-Gauss fit, trigg.
L MPV = 57.83 ± 0.19
Max = 63.79 ± 0.21
χ²/ndf = 37.27
SNR = 5.37
r:1 c:3 m:1 layer 2
Total events: 206862
Landau-Gauss fit, trigg.
L MPV = 52.42 ± 0.18
Max = 57.96 ± 0.20
χ²/ndf = 37.18
SNR = 6.34
μ-beam, E_b = 5 GeV
Run 380, V_{op} = 43.0 V
r:0 c:0 m:1
Total events: 524514
Landau-Gauss fit, trigg.
L MPV = 53.14 ± 0.21
Max = 58.43 ± 0.23
χ²/ndf = 17.71
SNR = 3.90
r:0 c:1 m:1
Total events: 540822
Landau-Gauss fit, trigg.
L MPV = 51.57 ± 0.24
Max = 57.59 ± 0.27
χ²/ndf = 25.66
SNR = 4.21
r:0 c:2 m:1
Total events: 542188
Landau-Gauss fit, trigg.
L MPV = 48.02 ± 0.19
Max = 53.56 ± 0.21
χ²/ndf = 17.47
SNR = 4.15
r:0 c:3 m:1
Total events: 543371
Landau-Gauss fit, trigg.
L MPV = 55.39 ± 0.27
Max = 61.29 ± 0.30
χ²/ndf = 15.91
SNR = 3.78
r:1 c:0 m:0
Total events: 752793
Landau-Gauss fit, trigg.
L MPV = 53.96 ± 0.21
Max = 59.47 ± 0.24
χ²/ndf = 19.66
SNR = 4.90
r:1 c:1 m:0
Total events: 770203
Landau-Gauss fit, trigg.
L MPV = 49.97 ± 0.15
Max = 55.01 ± 0.17
χ²/ndf = 27.21
SNR = 5.27
r:1 c:2 m:0
Total events: 749784
Landau-Gauss fit, trigg.
L MPV = 52.02 ± 0.18
Max = 57.54 ± 0.20
χ²/ndf = 25.37
SNR = 4.57
r:1 c:3 m:0
Total events: 621023
Landau-Gauss fit, trigg.
L MPV = 48.49 ± 0.37
Max = 55.82 ± 0.42
χ²/ndf = 17.23
SNR = 2.99
r:0 c:0 m:0
Total events: 771019
Landau-Gauss fit, trigg.
L MPV = 53.27 ± 0.25
Max = 58.65 ± 0.27
χ²/ndf = 13.67
SNR = 5.68
r:0 c:1 m:0
Total events: 771855
Landau-Gauss fit, trigg.
L MPV = 52.58 ± 0.18
Max = 57.58 ± 0.20
χ²/ndf = 18.25
SNR = 4.39
r:0 c:2 m:0
Total events: 772410
Landau-Gauss fit, trigg.
L MPV = 49.31 ± 0.19
Max = 54.06 ± 0.21
χ²/ndf = 15.03
SNR = 4.18
r:0 c:3 m:0
Total events: 772903
Landau-Gauss fit, trigg.
L MPV = 52.17 ± 0.26
Max = 57.32 ± 0.28
χ²/ndf = 11.72
SNR = 4.56
counts
ADC (arb. units)

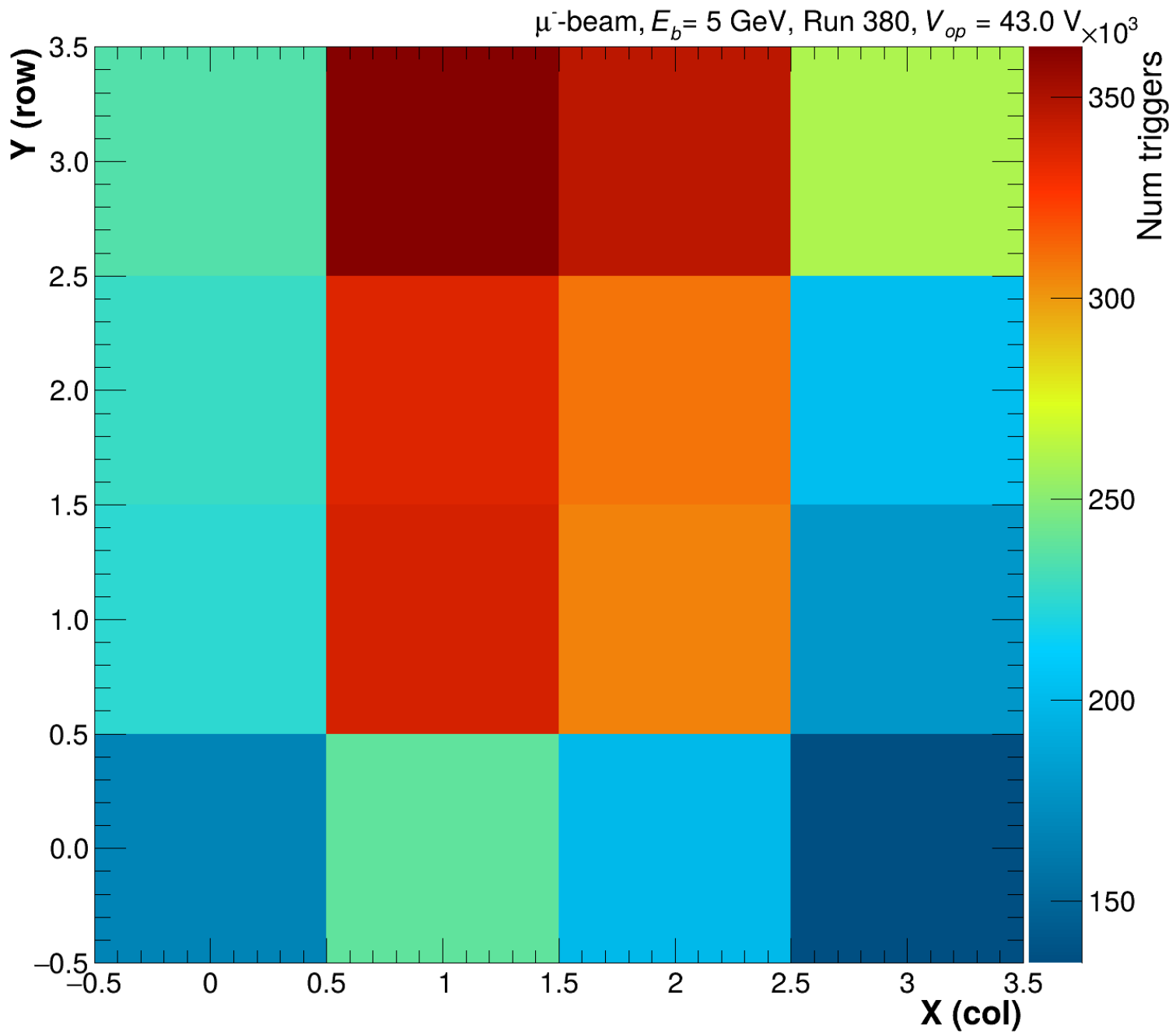

r:1 c:0 m:1
Total events: 536389
Landau-Gauss fit, trigg.
L MPV = 55.38 ± 0.26
Max = 60.56 ± 0.29
χ²/ndf = 14.43
SNR = 1.98
r:1 c:1 m:1
Total events: 294054
Landau-Gauss fit, trigg.
L MPV = 56.36 ± 0.26
Max = 62.61 ± 0.29
χ²/ndf = 29.45
SNR = 1.42
r:1 c:2 m:1
Total events: 542261
Landau-Gauss fit, trigg.
L MPV = 56.87 ± 0.25
Max = 62.71 ± 0.28
χ²/ndf = 14.76
SNR = 0.60
r:1 c:3 m:1 layer 3
Total events: 547946
Landau-Gauss fit, trigg.
L MPV = 53.40 ± 0.49
Max = 59.88 ± 0.55
χ²/ndf = 22.03
SNR = 0.77
μ⁻-beam, E_b = 5 GeV
Run 380, V_op = 43.0 V
r:0 c:0 m:1
Total events: 538087
Landau-Gauss fit, trigg.
L MPV = 54.31 ± 0.26
Max = 59.42 ± 0.29
χ²/ndf = 12.87
SNR = 0.60
r:0 c:1 m:1
Total events: 527646
Landau-Gauss fit, trigg.
L MPV = 55.35 ± 0.34
Max = 61.93 ± 0.38
χ²/ndf = 13.21
SNR = 0.67
r:0 c:2 m:1
Total events: 484041
Landau-Gauss fit, trigg.
L MPV = 51.94 ± 0.32
Max = 58.43 ± 0.36
χ²/ndf = 15.66
SNR = 0.63
r:0 c:3 m:1
Total events: 541306
Landau-Gauss fit, trigg.
L MPV = 59.34 ± 0.34
Max = 65.11 ± 0.38
χ²/ndf = 8.85
SNR = 0.49
r:1 c:0 m:0
Total events: 773582
Landau-Gauss fit, trigg.
L MPV = 56.07 ± 0.26
Max = 61.37 ± 0.29
χ²/ndf = 18.62
SNR = 1.82
r:1 c:1 m:0
Total events: 664716
Landau-Gauss fit, trigg.
L MPV = 55.25 ± 0.24
Max = 60.93 ± 0.26
χ²/ndf = 36.92
SNR = 1.38
r:1 c:2 m:0
Total events: 774415
Landau-Gauss fit, trigg.
L MPV = 56.43 ± 0.24
Max = 61.98 ± 0.26
χ²/ndf = 14.16
SNR = 0.60
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 773688
Landau-Gauss fit, trigg.
L MPV = 51.71 ± 0.27
Max = 56.57 ± 0.30
χ²/ndf = 11.74
SNR = 2.74
r:0 c:1 m:0
Total events: 773085
Landau-Gauss fit, trigg.
L MPV = 57.47 ± 0.32
Max = 63.54 ± 0.35
χ²/ndf = 10.87
SNR = 0.87
r:0 c:2 m:0
Total events: 754258
Landau-Gauss fit, trigg.
L MPV = 56.67 ± 0.32
Max = 62.69 ± 0.36
χ²/ndf = 11.34
SNR = 1.25
r:0 c:3 m:0
Total events: 774473
Landau-Gauss fit, trigg.
L MPV = 54.95 ± 0.33
Max = 60.30 ± 0.36
χ²/ndf = 6.37
SNR = 0.56
counts
ADC (arb. units)

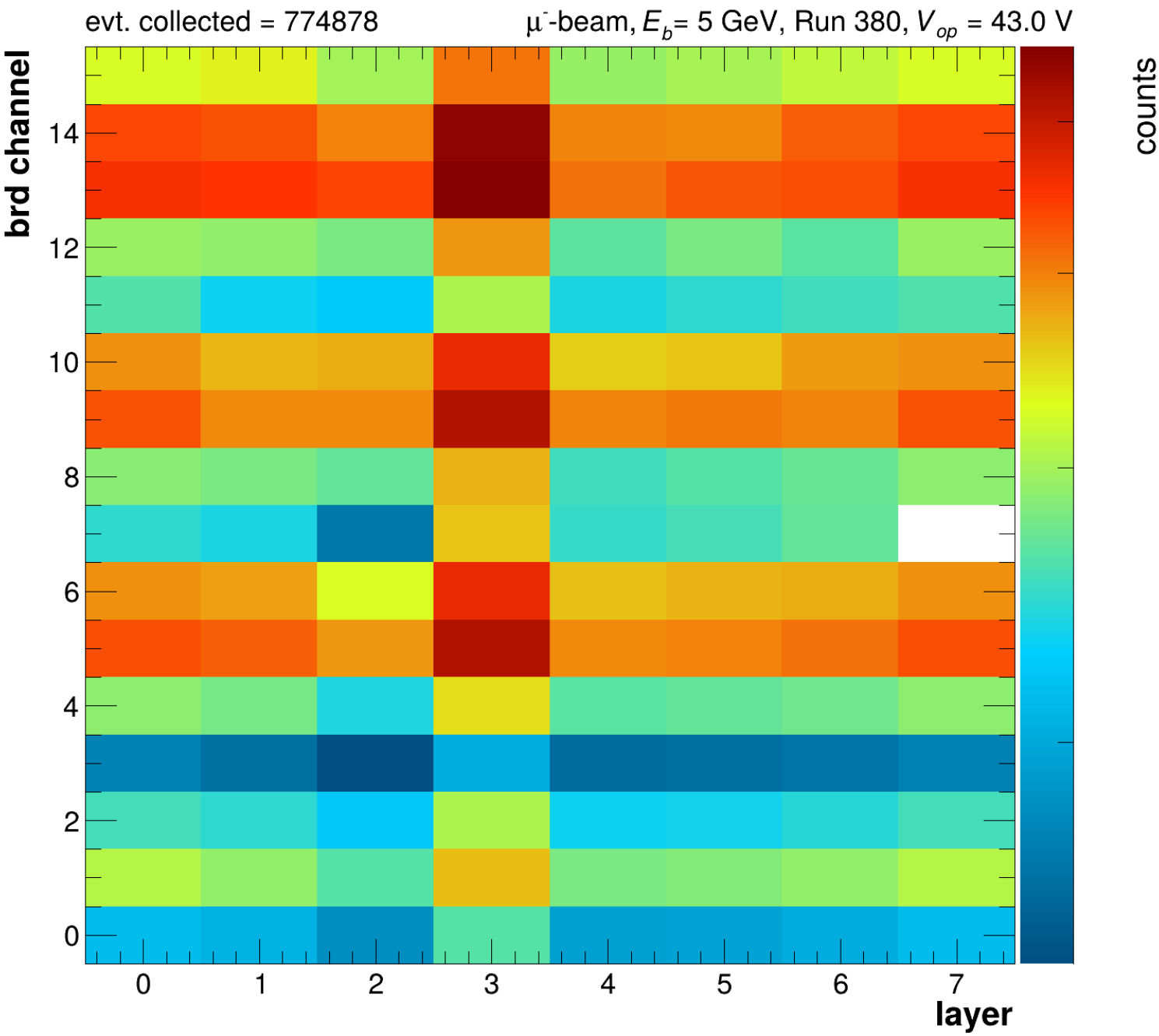

r:1 c:0 m:1
Total events: 547232
Landau-Gauss fit, trigg.
L MPV = 53.49 ± 0.18
Max = 57.93 ± 0.20
χ²/ndf = 14.56
SNR = 2.54
r:1 c:1 m:1
Total events: 545090
Landau-Gauss fit, trigg.
L MPV = 57.68 ± 0.17
Max = 62.70 ± 0.19
χ²/ndf = 22.23
SNR = 2.33
r:1 c:2 m:1
Total events: 542016
Landau-Gauss fit, trigg.
L MPV = 52.06 ± 0.15
Max = 56.60 ± 0.17
χ²/ndf = 14.53
SNR = 2.24
r:1 c:3 m:1 layer 4
Total events: 531508
Landau-Gauss fit, trigg.
L MPV = 60.55 ± 0.24
Max = 66.17 ± 0.26
χ²/ndf = 18.14
SNR = 2.29
μ⁻-beam, E_b = 5 GeV
Run 380, V_op = 43.0 V
r:0 c:0 m:1
Total events: 532979
Landau-Gauss fit, trigg.
L MPV = 60.82 ± 0.26
Max = 66.03 ± 0.29
χ²/ndf = 14.72
SNR = 2.11
r:0 c:1 m:1
Total events: 546295
Landau-Gauss fit, trigg.
L MPV = 52.24 ± 0.17
Max = 56.87 ± 0.18
χ²/ndf = 19.26
SNR = 1.97
r:0 c:2 m:1
Total events: 530381
Landau-Gauss fit, trigg.
L MPV = 61.07 ± 0.23
Max = 66.47 ± 0.25
χ²/ndf = 18.87
SNR = 1.99
r:0 c:3 m:1
Total events: 522561
Landau-Gauss fit, trigg.
L MPV = 59.52 ± 0.25
Max = 64.83 ± 0.27
χ²/ndf = 11.31
SNR = 2.00
r:1 c:0 m:0
Total events: 773449
Landau-Gauss fit, trigg.
L MPV = 55.94 ± 0.21
Max = 60.92 ± 0.23
χ²/ndf = 16.13
SNR = 2.33
r:1 c:1 m:0
Total events: 772566
Landau-Gauss fit, trigg.
L MPV = 55.96 ± 0.16
Max = 60.69 ± 0.17
χ²/ndf = 19.67
SNR = 2.36
r:1 c:2 m:0
Total events: 772882
Landau-Gauss fit, trigg.
L MPV = 53.56 ± 0.16
Max = 58.16 ± 0.17
χ²/ndf = 18.13
SNR = 2.38
r:1 c:3 m:0
Total events: 771884
Landau-Gauss fit, trigg.
L MPV = 57.47 ± 0.23
Max = 63.15 ± 0.26
χ²/ndf = 13.84
SNR = 1.89
r:0 c:0 m:0
Total events: 772827
Landau-Gauss fit, trigg.
L MPV = 55.94 ± 0.20
Max = 60.32 ± 0.21
χ²/ndf = 9.51
SNR = 2.57
r:0 c:1 m:0
Total events: 773396
Landau-Gauss fit, trigg.
L MPV = 58.00 ± 0.24
Max = 62.96 ± 0.26
χ²/ndf = 20.41
SNR = 2.31
r:0 c:2 m:0
Total events: 772208
Landau-Gauss fit, trigg.
L MPV = 58.19 ± 0.23
Max = 63.06 ± 0.24
χ²/ndf = 13.10
SNR = 2.56
r:0 c:3 m:0
Total events: 771551
Landau-Gauss fit, trigg.
L MPV = 57.55 ± 0.24
Max = 62.15 ± 0.26
χ²/ndf = 8.03
SNR = 2.64
counts
ADC (arb. units)

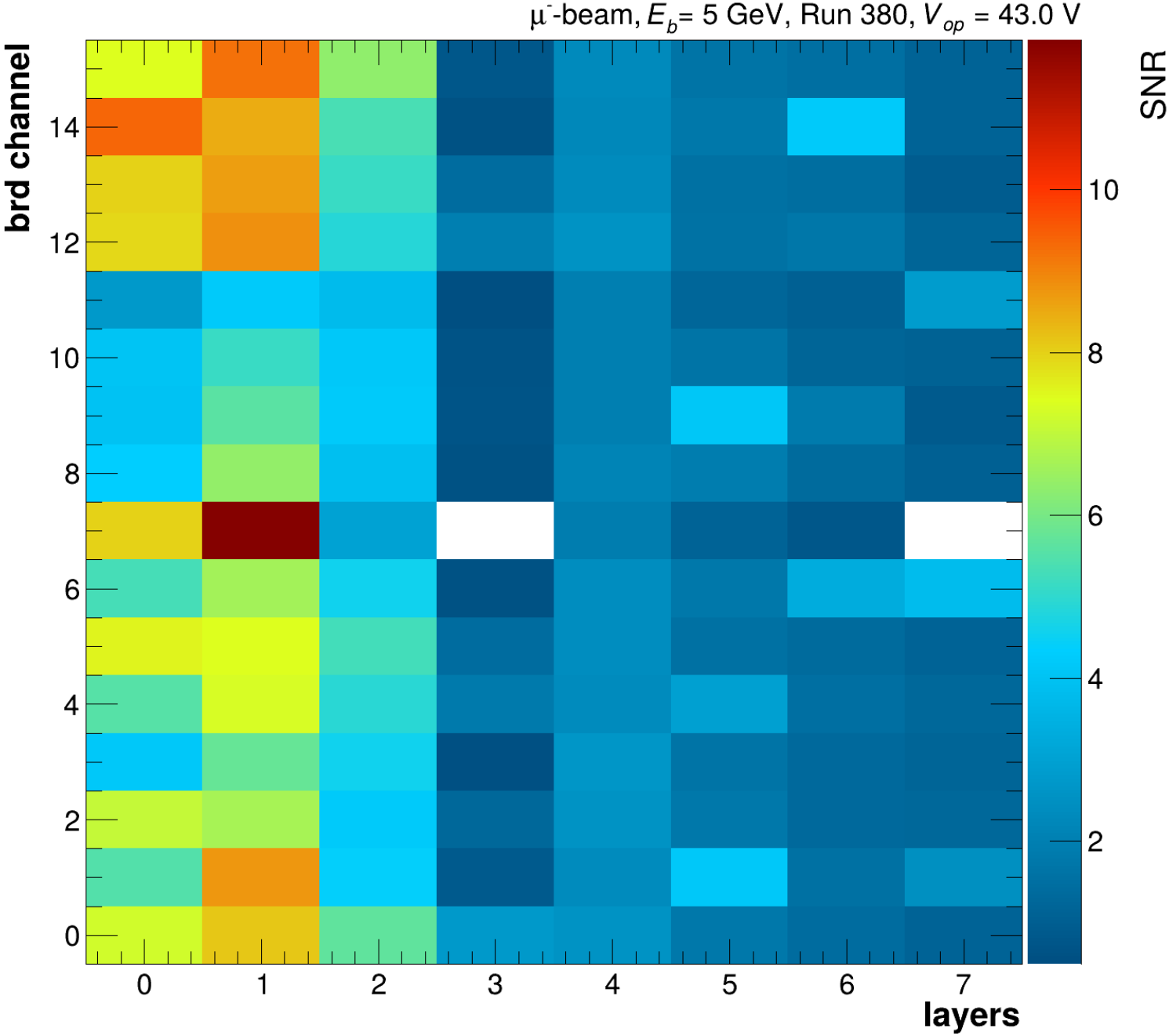

r:1 c:0 m:1
Total events: 538080
Landau-Gauss fit, trigg.
L MPV = 58.50 ± 0.16
Max = 62.14 ± 0.17
χ²/ndf = 11.24
SNR = 1.57
r:1 c:1 m:1
Total events: 538661
Landau-Gauss fit, trigg.
L MPV = 56.51 ± 0.14
Max = 60.58 ± 0.15
χ²/ndf = 19.30
SNR = 1.52
r:1 c:2 m:1
Total events: 484314
Landau-Gauss fit, trigg.
L MPV = 67.32 ± 0.19
Max = 72.55 ± 0.21
χ²/ndf = 21.36
SNR = 1.75
r:1 c:3 m:1 layer 5
Total events: 531980
Landau-Gauss fit, trigg.
L MPV = 62.74 ± 0.23
Max = 67.76 ± 0.25
χ²/ndf = 17.60
SNR = 1.63
μ⁻-beam, E_b = 5 GeV
Run 380, V_op = 43.0 V
r:0 c:0 m:1
Total events: 548275
Landau-Gauss fit, trigg.
L MPV = 61.15 ± 0.19
Max = 65.46 ± 0.21
χ²/ndf = 8.96
SNR = 1.89
r:0 c:1 m:1
Total events: 536401
Landau-Gauss fit, trigg.
L MPV = 59.72 ± 0.14
Max = 63.64 ± 0.15
χ²/ndf = 18.79
SNR = 4.10
r:0 c:2 m:1
Total events: 529743
Landau-Gauss fit, trigg.
L MPV = 68.67 ± 0.25
Max = 73.94 ± 0.27
χ²/ndf = 20.67
SNR = 1.61
r:0 c:3 m:1
Total events: 542420
Landau-Gauss fit, trigg.
L MPV = 63.75 ± 0.21
Max = 68.05 ± 0.22
χ²/ndf = 4.96
SNR = 1.23
r:1 c:0 m:0
Total events: 773032
Landau-Gauss fit, trigg.
L MPV = 65.46 ± 0.20
Max = 70.09 ± 0.22
χ²/ndf = 14.78
SNR = 2.97
r:1 c:1 m:0
Total events: 772659
Landau-Gauss fit, trigg.
L MPV = 68.11 ± 0.20
Max = 73.27 ± 0.22
χ²/ndf = 23.85
SNR = 1.53
r:1 c:2 m:0
Total events: 753686
Landau-Gauss fit, trigg.
L MPV = 55.56 ± 0.15
Max = 59.67 ± 0.16
χ²/ndf = 14.27
SNR = 1.74
r:1 c:3 m:0
Total events: 772657
Landau-Gauss fit, trigg.
L MPV = 62.53 ± 0.30
Max = 68.44 ± 0.33
χ²/ndf = 10.44
SNR = 1.13
r:0 c:0 m:0
Total events: 774332
Landau-Gauss fit, trigg.
L MPV = 65.10 ± 0.27
Max = 70.12 ± 0.29
χ²/ndf = 11.41
SNR = 1.77
r:0 c:1 m:0
Total events: 773049
Landau-Gauss fit, trigg.
L MPV = 65.83 ± 0.20
Max = 70.43 ± 0.21
χ²/ndf = 19.41
SNR = 4.17
r:0 c:2 m:0
Total events: 772724
Landau-Gauss fit, trigg.
L MPV = 65.38 ± 0.25
Max = 70.30 ± 0.27
χ²/ndf = 12.07
SNR = 1.78
r:0 c:3 m:0
Total events: 774229
Landau-Gauss fit, trigg.
L MPV = 63.52 ± 0.29
Max = 68.42 ± 0.31
χ²/ndf = 9.82
SNR = 1.62
counts
ADC (arb. units)

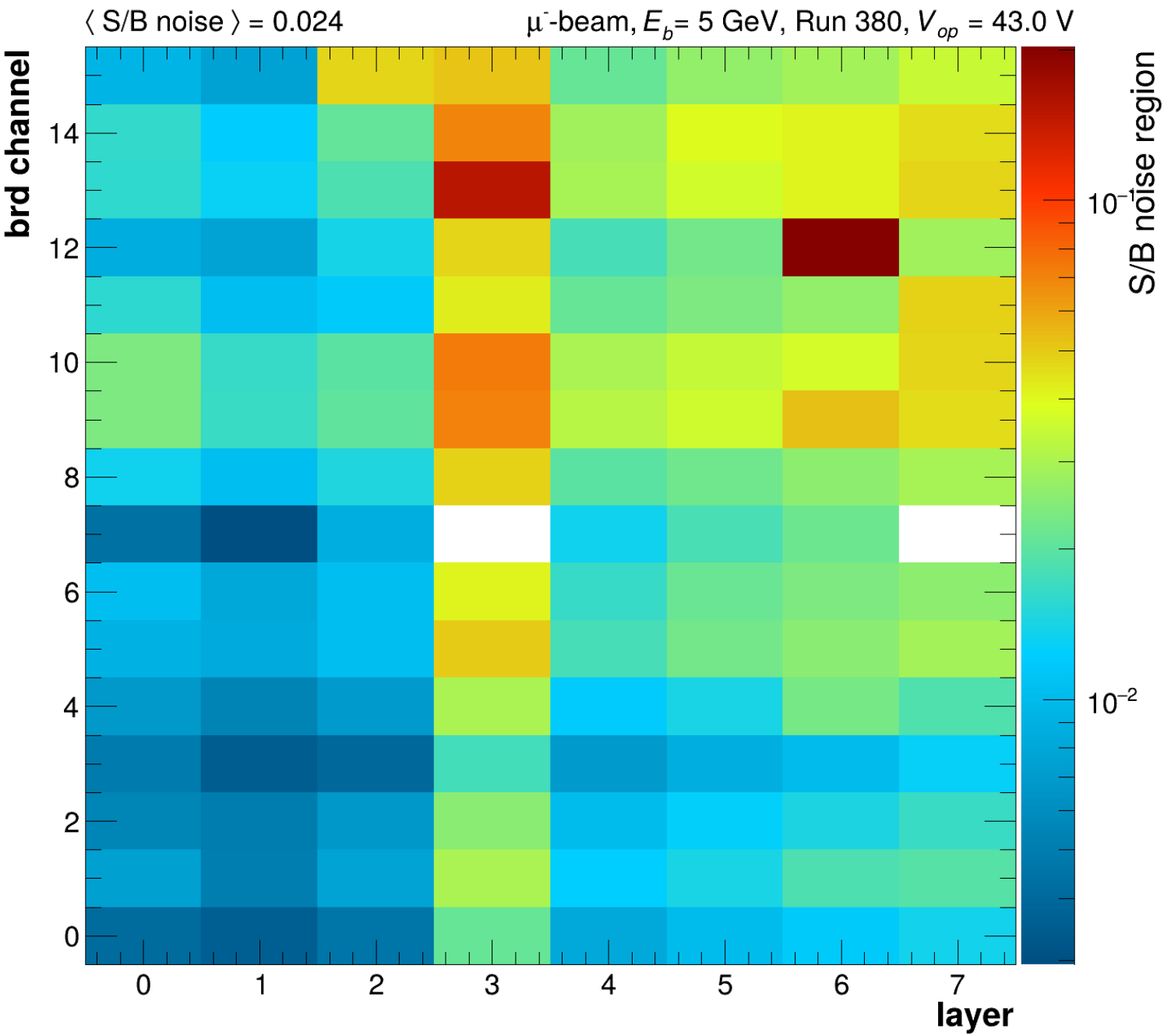

r:1 c:0 m:1
Total events: 117616
Landau-Gauss fit, trigg.
L MPV = 76.70 ± 0.30
Max = 82.53 ± 0.33
χ²/ndf = 19.84
SNR = 1.70
r:1 c:1 m:1
Total events: 541014
Landau-Gauss fit, trigg.
L MPV = 73.90 ± 0.18
Max = 78.89 ± 0.19
χ²/ndf = 10.57
SNR = 1.42
r:1 c:2 m:1
Total events: 534000
Landau-Gauss fit, trigg.
L MPV = 67.70 ± 0.19
Max = 72.61 ± 0.20
χ²/ndf = 24.11
SNR = 4.20
r:1 c:3 m:1 layer 6
Total events: 541519
Landau-Gauss fit, trigg.
L MPV = 77.28 ± 0.24
Max = 82.66 ± 0.25
χ²/ndf = 8.19
SNR = 1.48
μ⁻-beam, E_b = 5 GeV
Run 380, V_op = 43.0 V
r:0 c:0 m:1
Total events: 547741
Landau-Gauss fit, trigg.
L MPV = 69.64 ± 0.34
Max = 75.15 ± 0.37
χ²/ndf = 11.28
SNR = 1.33
r:0 c:1 m:1
Total events: 410097
Landau-Gauss fit, trigg.
L MPV = 78.13 ± 0.28
Max = 83.80 ± 0.30
χ²/ndf = 26.45
SNR = 1.84
r:0 c:2 m:1
Total events: 538511
Landau-Gauss fit, trigg.
L MPV = 70.68 ± 0.23
Max = 75.79 ± 0.24
χ²/ndf = 15.78
SNR = 1.17
r:0 c:3 m:1
Total events: 541495
Landau-Gauss fit, trigg.
L MPV = 69.65 ± 0.24
Max = 74.47 ± 0.26
χ²/ndf = 5.76
SNR = 1.03
r:1 c:0 m:0
Total events: 587875
Landau-Gauss fit, trigg.
L MPV = 68.03 ± 0.24
Max = 73.02 ± 0.26
χ²/ndf = 16.72
SNR = 1.50
r:1 c:1 m:0
Total events: 773960
Landau-Gauss fit, trigg.
L MPV = 72.87 ± 0.20
Max = 78.09 ± 0.21
χ²/ndf = 16.39
SNR = 1.36
r:1 c:2 m:0
Total events: 773159
Landau-Gauss fit, trigg.
L MPV = 70.59 ± 0.17
Max = 75.09 ± 0.18
χ²/ndf = 20.33
SNR = 3.29
r:1 c:3 m:0
Total events: 774246
Landau-Gauss fit, trigg.
L MPV = 66.03 ± 0.36
Max = 71.60 ± 0.39
χ²/ndf = 9.07
SNR = 0.77
r:0 c:0 m:0
Total events: 774546
Landau-Gauss fit, trigg.
L MPV = 70.02 ± 0.29
Max = 75.10 ± 0.31
χ²/ndf = 7.89
SNR = 1.40
r:0 c:1 m:0
Total events: 717337
Landau-Gauss fit, trigg.
L MPV = 67.49 ± 0.22
Max = 72.46 ± 0.23
χ²/ndf = 20.18
SNR = 1.55
r:0 c:2 m:0
Total events: 773496
Landau-Gauss fit, trigg.
L MPV = 70.53 ± 0.21
Max = 75.15 ± 0.22
χ²/ndf = 8.40
SNR = 1.33
r:0 c:3 m:0
Total events: 774359
Landau-Gauss fit, trigg.
L MPV = 73.24 ± 0.30
Max = 78.43 ± 0.32
χ²/ndf = 6.59
SNR = 1.31
counts
ADC (arb. units)

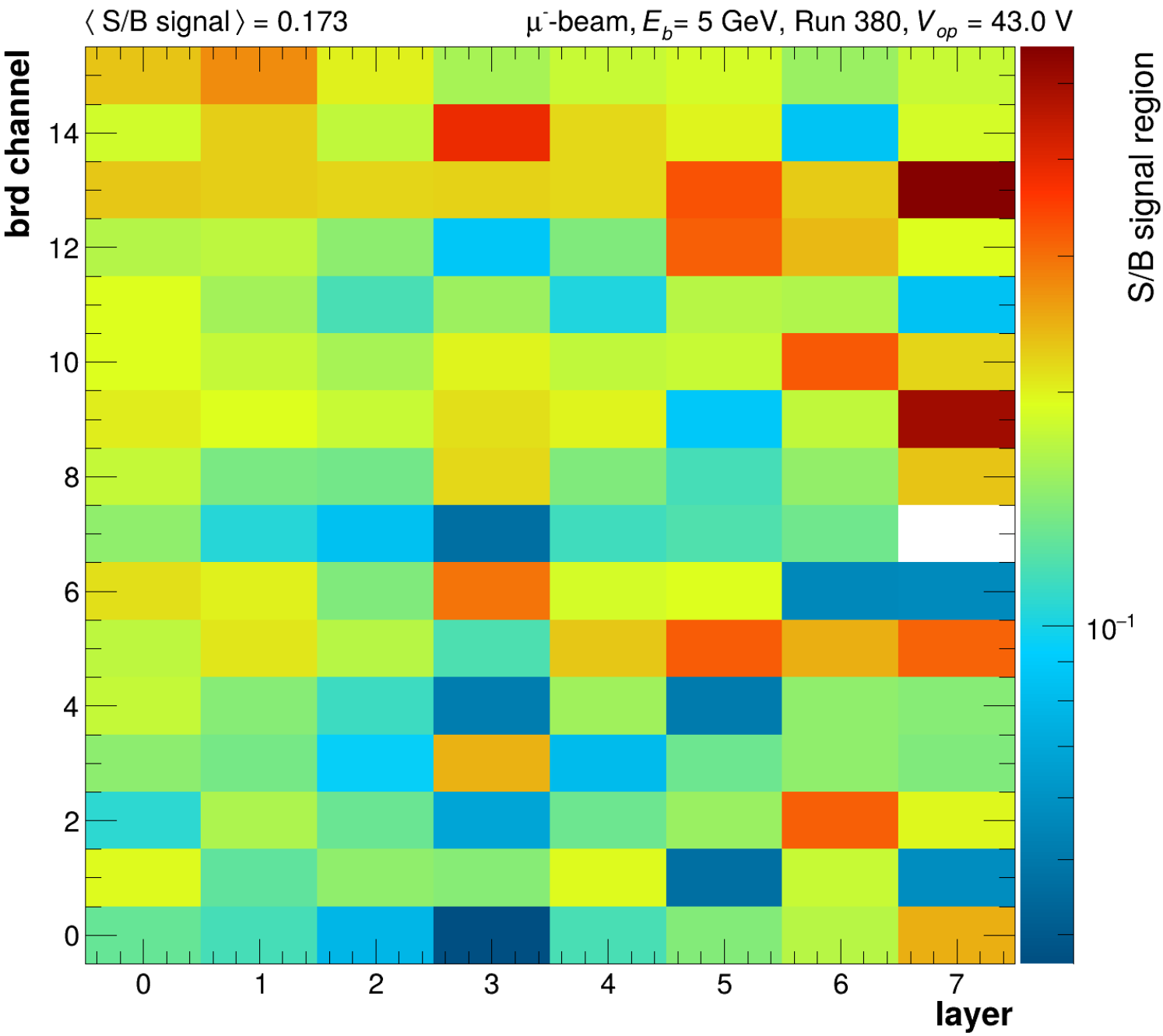

r:1 c:0 m:1
Total events: 547944
Landau-Gauss fit, trigg.
L MPV = 52.33 ± 0.30
Max = 58.19 ± 0.34
χ²/ndf = 12.39
SNR = 1.19
r:1 c:1 m:1
Total events: 538384
Landau-Gauss fit, trigg.
L MPV = 55.00 ± 0.16
Max = 59.38 ± 0.17
χ²/ndf = 16.52
SNR = 0.92
r:1 c:2 m:1
Total events: 534225
Landau-Gauss fit, trigg.
L MPV = 52.81 ± 0.20
Max = 57.89 ± 0.22
χ²/ndf = 23.57
SNR = 1.15
r:1 c:3 m:1 layer 7
Total events: 529191
Landau-Gauss fit, trigg.
L MPV = 50.76 ± 0.21
Max = 55.80 ± 0.24
χ²/ndf = 10.33
SNR = 1.13
μ⁻-beam, E_b = 5 GeV
Run 380, V_op = 43.0 V
r:0 c:0 m:1
Total events: 542407
Landau-Gauss fit, trigg.
L MPV = 53.40 ± 0.20
Max = 57.54 ± 0.22
χ²/ndf = 5.99
SNR = 1.03
r:0 c:1 m:1
Total events: 538097
Landau-Gauss fit, trigg.
L MPV = 55.57 ± 0.17
Max = 60.13 ± 0.19
χ²/ndf = 15.27
SNR = 0.88
r:0 c:2 m:1
Total events: 484530
Landau-Gauss fit, trigg.
L MPV = 54.21 ± 0.20
Max = 59.52 ± 0.22
χ²/ndf = 18.11
SNR = 1.08
r:0 c:3 m:1
Total events: 343943
Landau-Gauss fit, trigg.
L MPV = 55.88 ± 0.26
Max = 61.12 ± 0.29
χ²/ndf = 11.73
SNR = 2.82
r:1 c:0 m:0
Total events: 774625
Landau-Gauss fit, trigg.
L MPV = 55.04 ± 0.22
Max = 60.14 ± 0.25
χ²/ndf = 8.85
SNR = 1.25
r:1 c:1 m:0
Total events: 773494
Landau-Gauss fit, trigg.
L MPV = 55.05 ± 0.16
Max = 59.73 ± 0.18
χ²/ndf = 17.24
SNR = 1.12
r:1 c:2 m:0
Total events: 773384
Landau-Gauss fit, trigg.
L MPV = 56.33 ± 0.16
Max = 60.76 ± 0.18
χ²/ndf = 19.16
SNR = 3.79
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 774536
Landau-Gauss fit, trigg.
L MPV = 51.34 ± 0.23
Max = 55.95 ± 0.25
χ²/ndf = 6.89
SNR = 1.13
r:0 c:1 m:0
Total events: 773578
Landau-Gauss fit, trigg.
L MPV = 56.60 ± 0.19
Max = 61.25 ± 0.21
χ²/ndf = 10.87
SNR = 2.45
r:0 c:2 m:0
Total events: 754271
Landau-Gauss fit, trigg.
L MPV = 55.47 ± 0.29
Max = 61.35 ± 0.32
χ²/ndf = 18.84
SNR = 1.25
r:0 c:3 m:0
Total events: 685739
Landau-Gauss fit, trigg.
L MPV = 53.15 ± 0.21
Max = 56.79 ± 0.23
χ²/ndf = 6.96
SNR = 1.17
counts
ADC (arb. units)

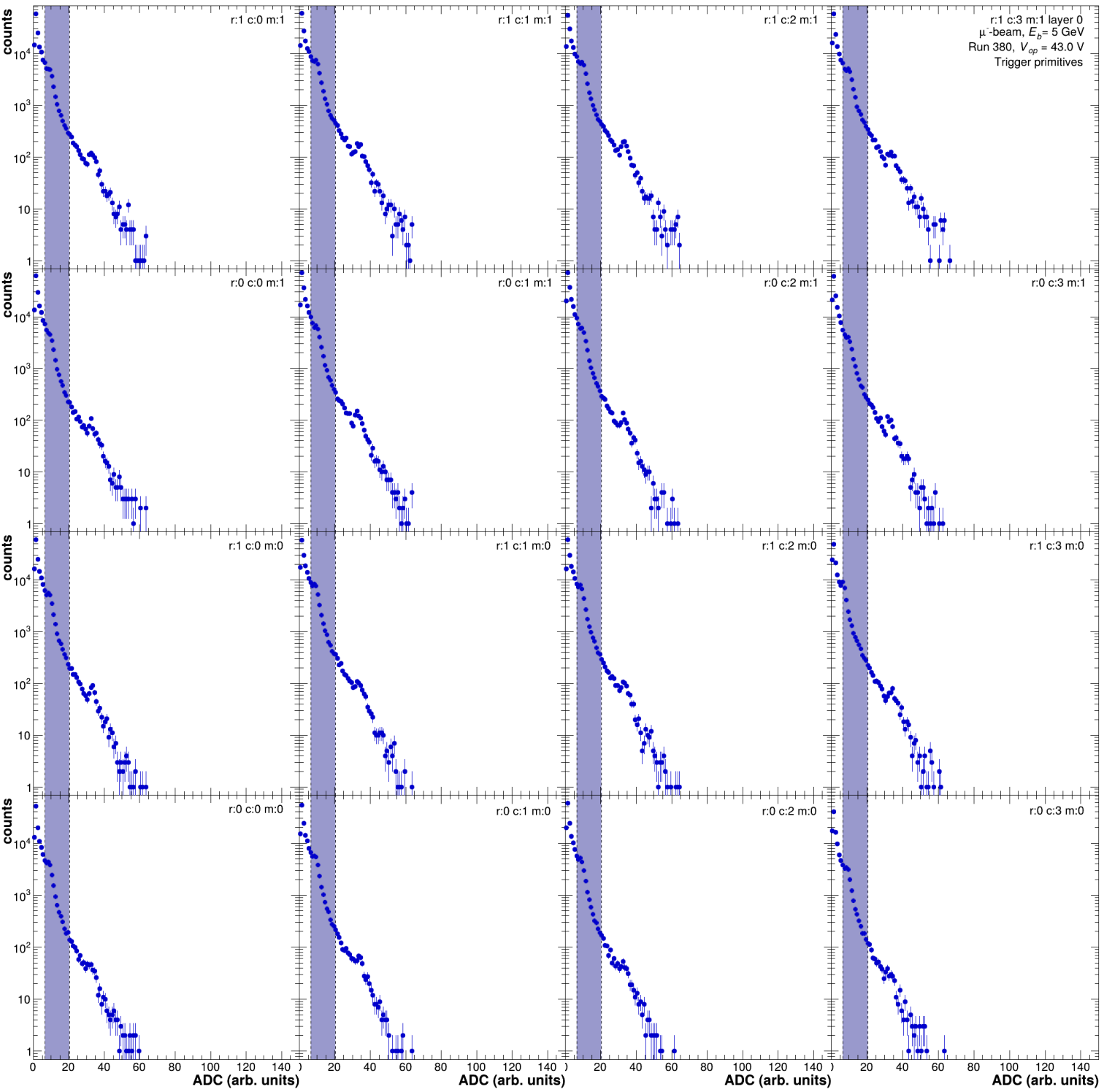

μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
Y (row)
X (col)
Num triggers
×10³
350
300
250
200
150
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5

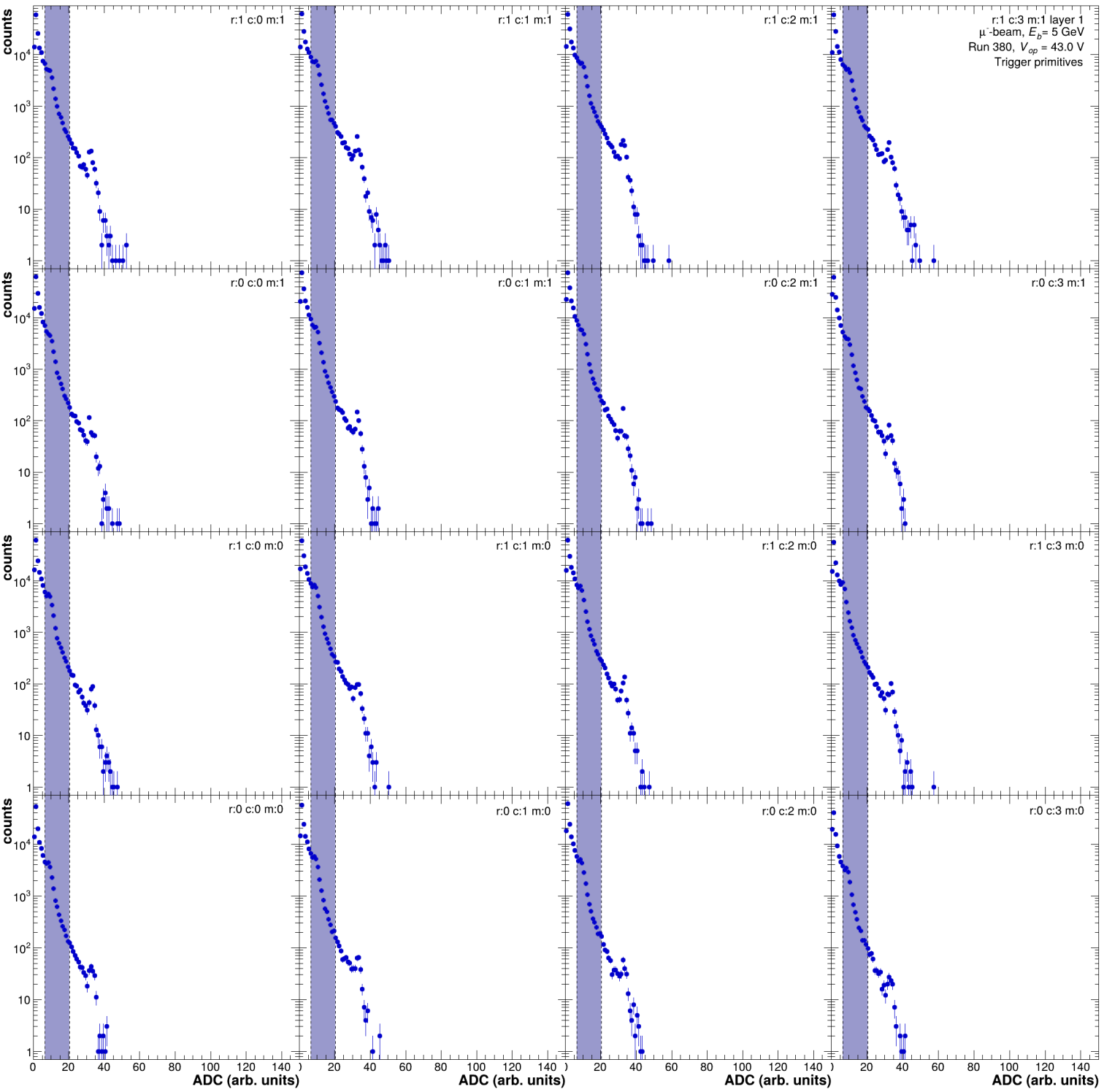

evt. collected = 774878
μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts

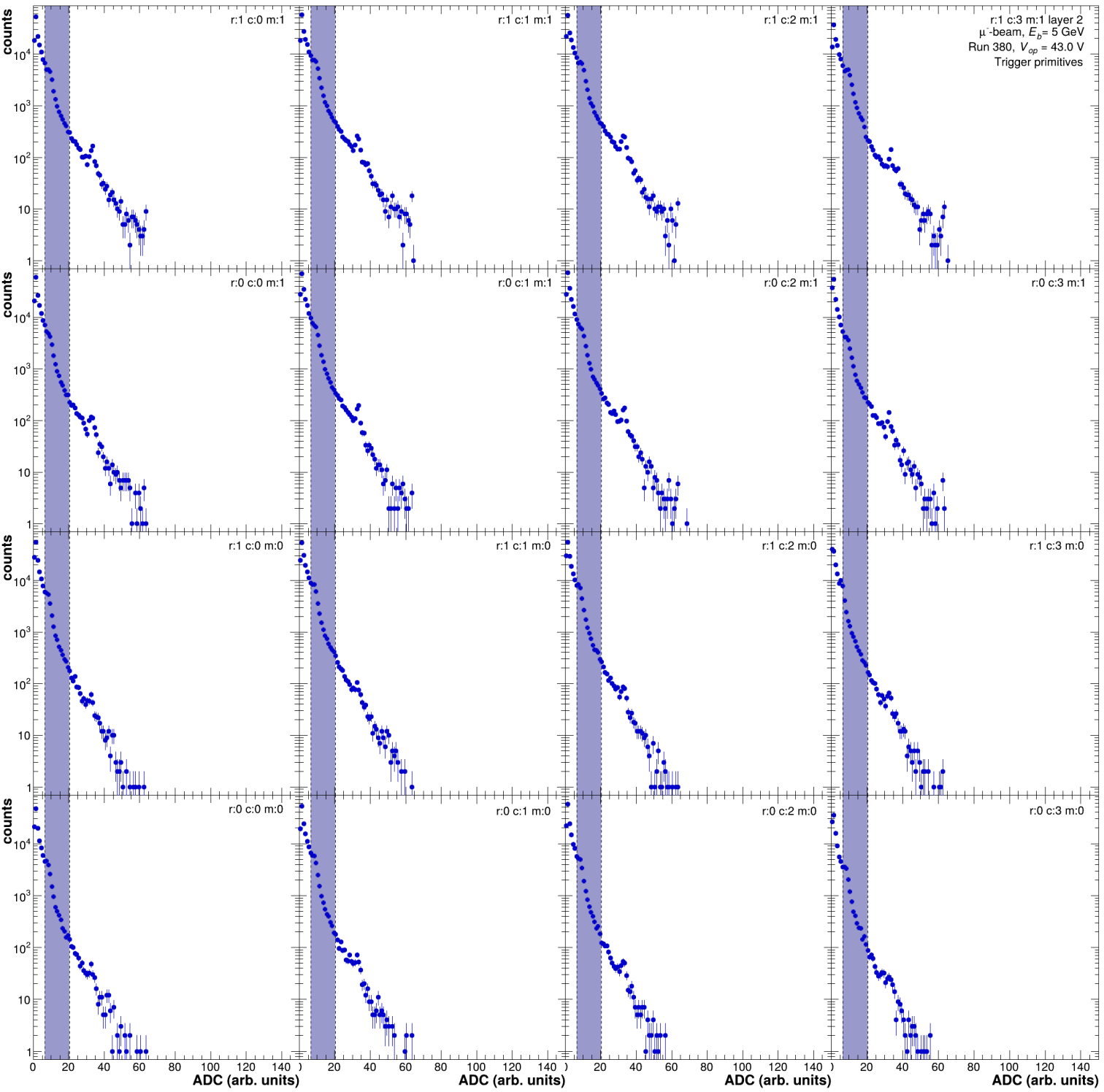

μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
0
2
4
6
8
10
12
14
layers
0
1
2
3
4
5
6
7
SNR
2
4
6
8
10

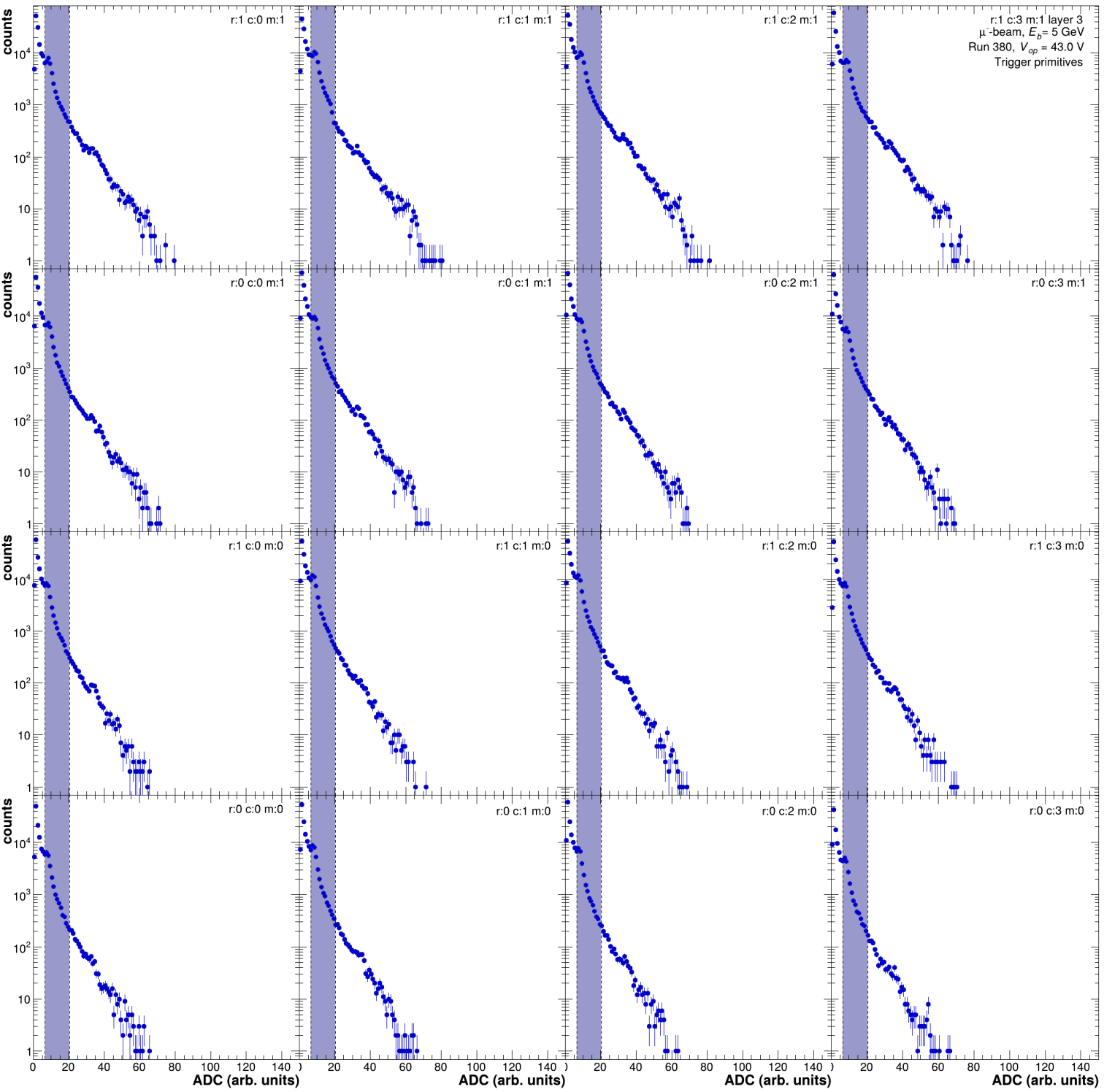

⟨ S/B noise ⟩ = 0.024
μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
layer
S/B noise region
10^{-1}
10^{-2}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

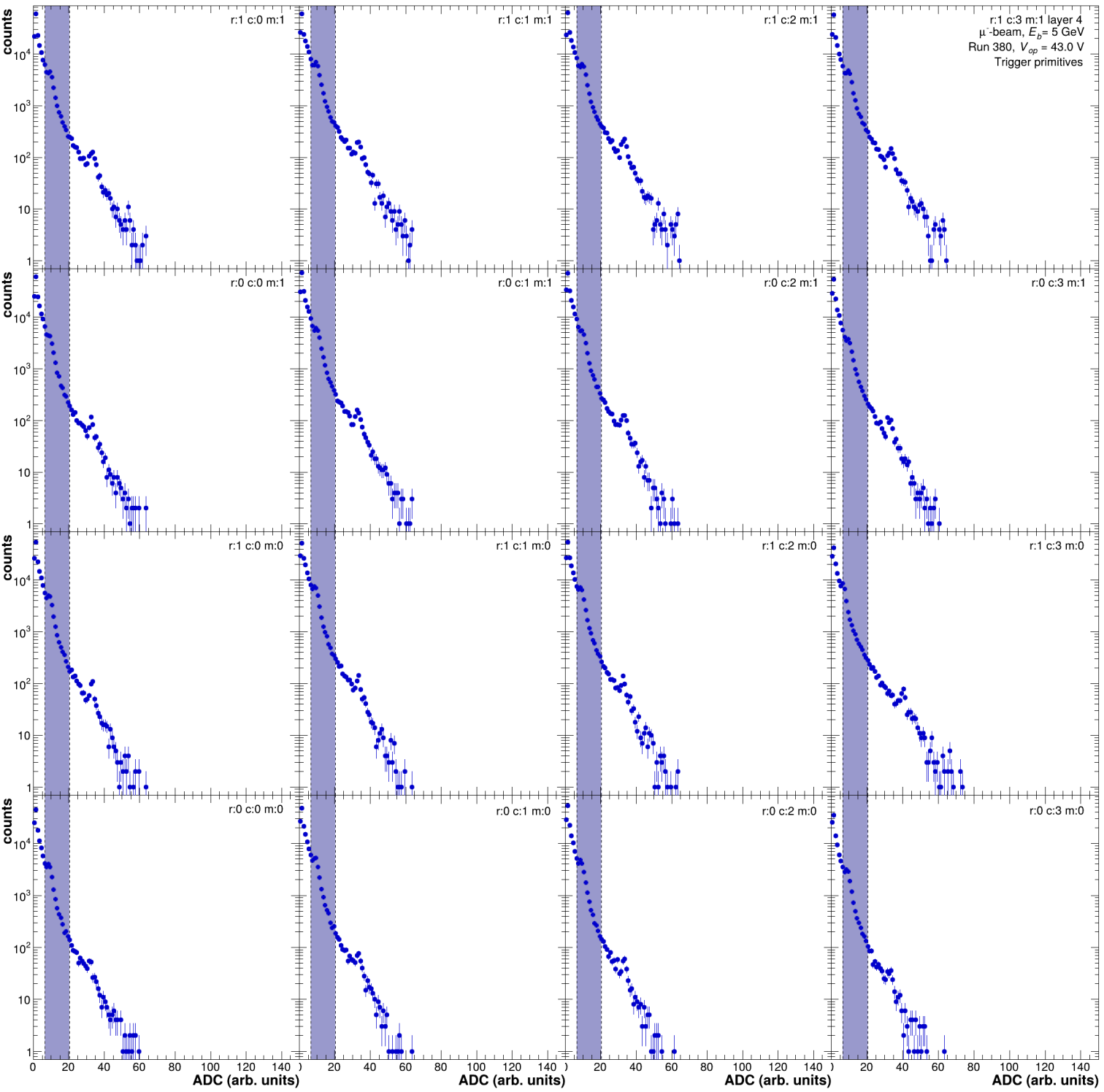

⟨ S/B signal ⟩ = 0.173
μ⁻-beam, E_b= 5 GeV, Run 380, V_{op} = 43.0 V
brd channel
layer
S/B signal region
10^{-1}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

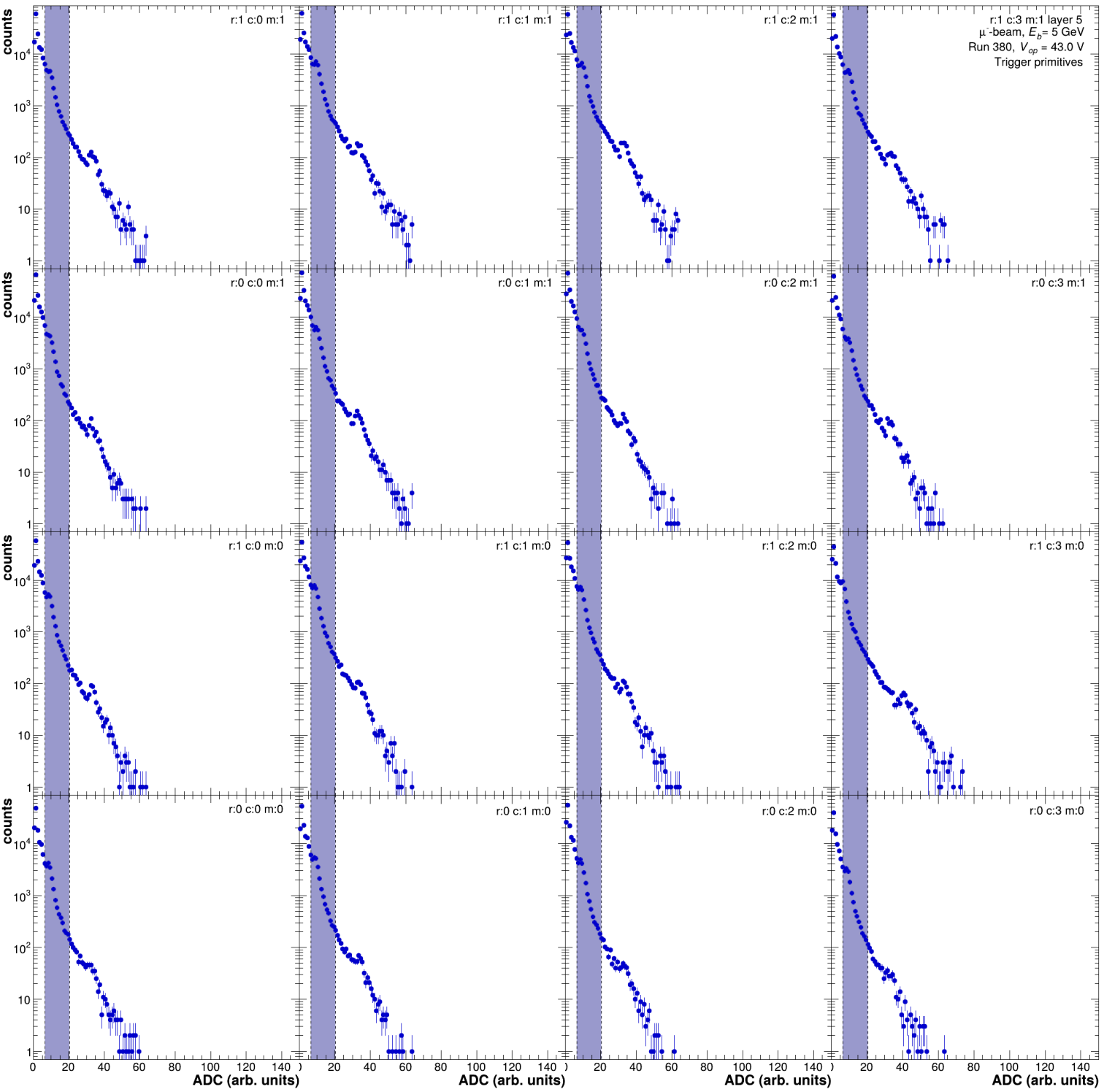

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

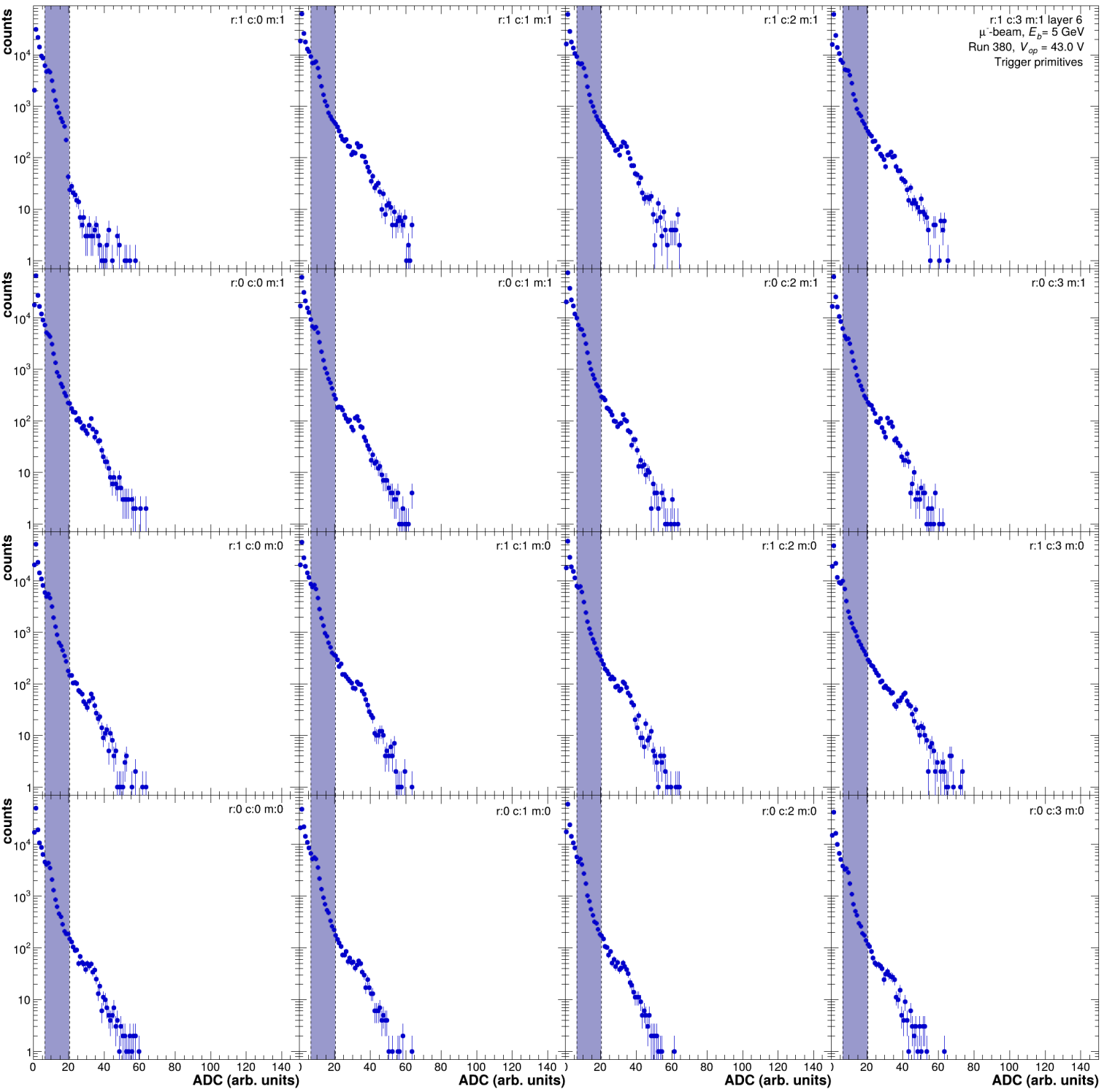

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

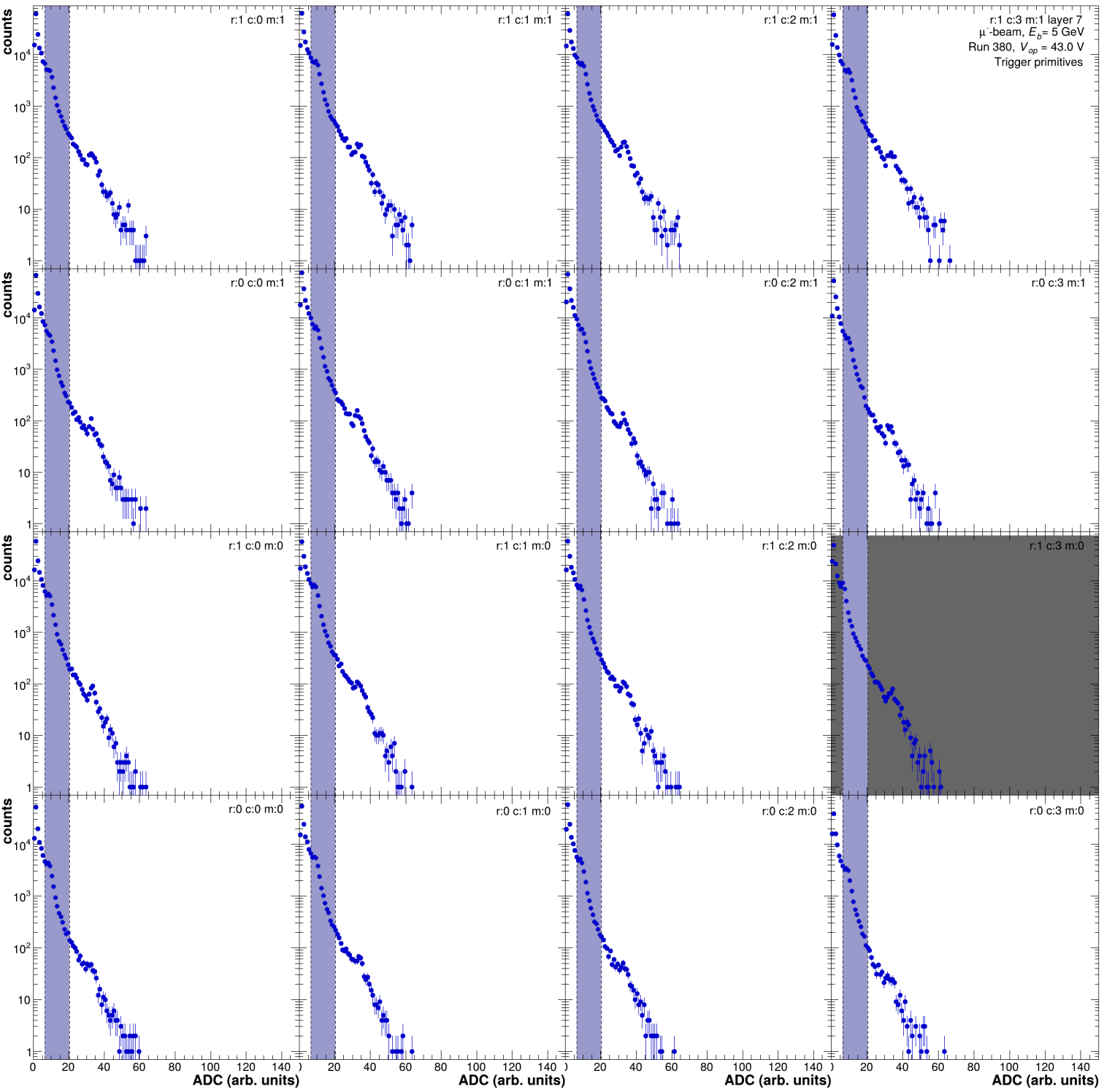

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 380, V_{op} = 43.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)